Revisiting Continual Learning in Language Models: A Three-Interface Diagnosis of Stability and Plasticity

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Abstract

Continual learning in language models is commonly studied in the incremental learning (IL) setting, where a model learns a sequence of tasks and is evaluated through its final classifier or language-model (LM) head. This interface, however, does not indicate where forgetting occurs: a drop in accuracy may reflect representation degradation in the backbone, readout mismatch in the head, or both. We therefore argue that backward transfer (BWT) and forward transfer (FWT) should be interpreted as interface-dependent quantities rather than as single model-level properties. We study three evaluation interfaces: the trained head, which is updated sequentially with the backbone; a probe newly trained on frozen backbone features; and a classifier-free backbone diagnostic. They measure deployed performance, recoverability under a new readout, and representation separability, respectively. Across class-incremental experiments with encoder and decoder language backbones, plus task-incremental sanity checks, conclusions about forgetting and plasticity change with the interface: the trained head often indicates severe forgetting, the probe near-complete recoverability, and the backbone diagnostic intermediate representation drift with a consistent negative relation between stability and plasticity. This relation holds for clustering-free diagnostics such as nearest-class-mean accuracy and across five clustering algorithms and three metrics. Finally, Just LM-Head Tuning (JLT), a head-only intervention on a fixed incrementally trained backbone, recovers 96–99% of the gap to joint training for replay- and distillation-based methods but only about three quarters for plain sequential fine-tuning, suggesting that many failures attributed to catastrophic forgetting reflect head–backbone mismatch plus partial representation drift. Our results call for reporting where forgetting is measured, not only how much. Code is available at https://github.com/kkt94/RevisitingCL.

1 Introduction

Continual learning studies how a model can keep acquiring new knowledge while retaining what it has already learned [Parisi et al., 2019]. A widely studied setting is incremental learning (IL), in which a model is trained on a sequence of tasks and evaluated on all tasks seen so far [Polikar et al., 2001, Kemker and Kanan, 2018]. In language modeling and NLP, this problem is especially important because pretrained language models (PLMs) are often adapted to new domains or task streams through sequential fine-tuning. A persistent challenge is catastrophic forgetting: performance on earlier tasks can degrade sharply after the model is updated on later tasks [French, 1999, Li et al., 2019, Hu et al., 2019, Kaushik et al., 2021, van de Ven et al., 2025]. Many IL methods therefore focus on memory stability through replay, regularization, or distillation [Kirkpatrick et al., 2017, Wang et al., 2023, Yang et al., 2025b].

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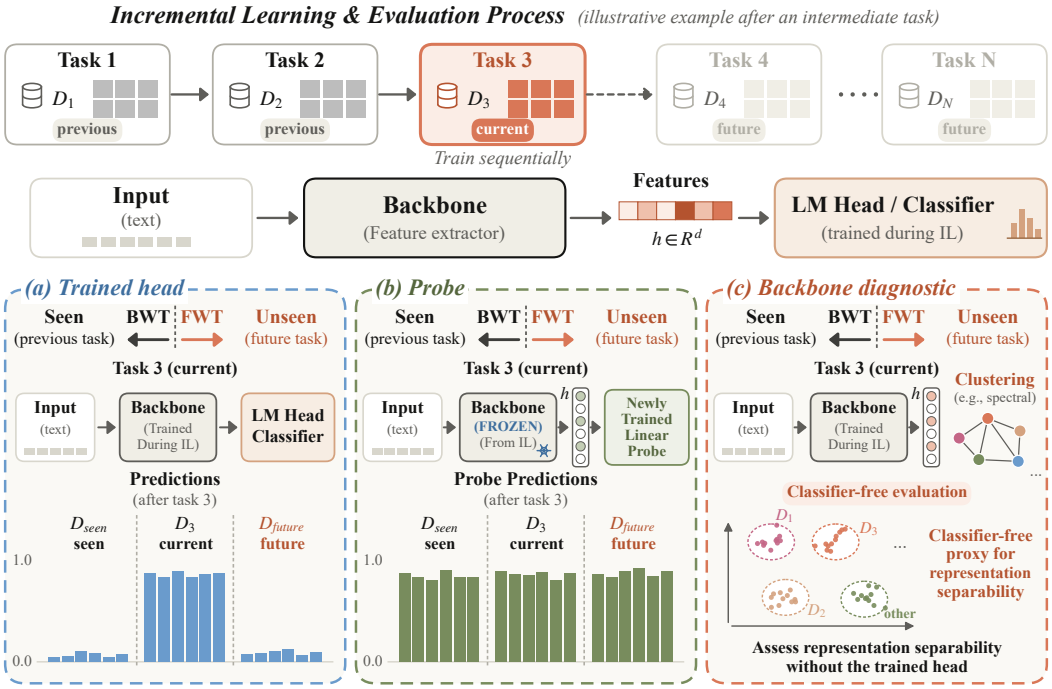


Figure 1: Overview of evaluating BWT and operational FWT at three interfaces in discriminative backbones. We measure transfer through (a) the trained head, (b) a probe, and (c) the backbone diagnostic, a classifier-free proxy for representation separability. The backbone diagnostic does not replace task accuracy; it evaluates separability without using the trained head. In generative backbones, the last-token hidden state of the prompt, from which the label is generated, replaces the [CLS] feature, and the LM head serves as the trained head. Details are in Appendix A.

However, standard IL evaluation usually observes the model through the final classifier or language-model (LM) head. This interface is useful for measuring deployed task accuracy, but it can also mix together two different phenomena: degradation of the backbone representation and failure of the prediction head to read out information that is still present in the backbone. This distinction matters for the *Stability–Plasticity Dilemma* [Abraham and Robins, 2005, Mermillod et al., 2013, Wu et al., 2021, Araujo et al., 2022]. If forgetting is mainly a head-level readout problem, then low head-level accuracy does not necessarily imply that the backbone has irreversibly lost the task. Conversely, if a newly trained probe compensates for weak or shifted representations, it may understate representational forgetting.

Recent studies suggest that the backbone can retain more information than head-level accuracy indicates, and that much of the observed degradation can arise in the classifier or LM head [Davari et al., 2022, Zheng et al., 2024, 2025a]. Other work studies Forward Transfer (FWT) and argues that less forgetting can provide a useful inductive bias for future tasks [Chen et al., 2023, Zheng et al., 2025b]. These findings raise a more basic measurement question: when an incremental learner fails on old tasks, where is forgetting being measured?

We address this question by revisiting Backward Transfer (BWT) and FWT under three evaluation interfaces: the trained head (the classifier or LM head updated sequentially with the backbone), a probe (a linear classifier newly trained on frozen backbone features), and the backbone diagnostic (classifier-free separability of backbone features). The main contribution is not a new metric that replaces accuracy. It is a diagnostic decomposition showing that standard IL metrics conflate three distinct estimands: deployed head behavior, recoverable information under a new readout, and classifier-free representation separability. Since the backbone has no standalone prediction interface, we approximate changes in backbone separability using clustering-based and clustering-free diagnostics.

Our analysis shows that the apparent relationship between BWT and FWT changes substantially across these interfaces. Head-level metrics often overstate forgetting because the head is itself

unstable under sequential updates, while the probe can understate forgetting because a newly trained classifier can compensate for shifted or weakened representations. The backbone diagnostic suggests intermediate representation drift and a consistent stability–plasticity pattern across clustering-free diagnostics and clustering choices.

Motivated by this gap between interfaces, we study Just LM-Head Tuning (JLT), a simple head-only adaptation procedure. JLT is not a new continual learning algorithm but a diagnostic intervention: by re-optimizing only the LM head on top of a fixed, incrementally trained backbone, we estimate how much performance can be recovered by correcting head–backbone mismatch. Because the head may be adapted before each task is evaluated, JLT scores should be read as recoverable performance, not as standard class-incremental learning (CIL) accuracy.

We evaluate this diagnostic view across five benchmark groups spanning seven datasets, encoder and decoder language backbones, multiple IL methods in CIL, and a discriminative-backbone sanity check in task-incremental learning (TIL). Concretely, we formalize BWT and an operational FWT (plasticity-gain) measure as interface-dependent quantities: the same formula answers different questions when measured through the trained head, a probe, or the backbone diagnostic. On this basis, we provide a three-interface diagnostic framework that separates deployed performance, readout recoverability, and classifier-free representation separability. Empirically, we show that these interfaces lead to qualitatively different conclusions, that the backbone-level stability–plasticity pattern is consistent across clustering-free diagnostics and clustering choices, and that JLT can be used as a head-only intervention to estimate how much apparent forgetting is recoverable through readout re-alignment and how much remains as a representation-level residual. Unlike prior probing studies, which examine stability only [Davari et al., 2022, Zheng et al., 2024, 2025a], this framing measures representation-level plasticity and compares the BWT–FWT relation across interfaces (Appendix Table 21).

2 Metrics are interface-dependent

2.1 Problem setup

We consider an incremental learner that receives a sequence of tasks $D = \{D_1, D_2, \cdots, D_T\}$, where each task $D_t = \{(x_t^i, y_t^i)\}_{i=1}^{N_t}$ contains inputs $x_t^i \in X_t$ and labels $y_t^i \in Y_t$. The learner updates the same model sequentially from task 1 to task T.

The two most common IL scenarios are CIL and TIL. In CIL, label sets from different tasks do not overlap, i.e., $Y_i \cap Y_j = \emptyset$ for $i \neq j$, and the model must predict among all classes observed so far. In TIL, the task identity is available at evaluation time, so prediction is restricted to the classes of the corresponding task. Because CIL requires a shared prediction space across tasks, it is usually more challenging than TIL [Tao et al., 2023]. We focus on CIL in the main results and use TIL as a sanity check in Appendix D.

2.2 Evaluation metrics for IL

BWT [Lopez-Paz and Ranzato, 2017, Ebrahimi et al., 2019, 2020] is a representative IL metric for *memory stability*. It compares performance on a previous task immediately after that task is learned with performance on the same task after the full task sequence has been learned:

$$BWT = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{T,i} - a_{i,i}) \tag{1}$$

where T is the last task, $a_{T,i}$ is the score on the i-th task after training up to the T-th task, and $a_{i,i}$ is the score on the i-th task immediately after training that task. The score a can be instantiated at three evaluation interfaces: the trained head, a probe, or the backbone diagnostic (Section 2.3). We denote the resulting metrics BWT-head, BWT-probe, and BWT-backbone, and likewise for FWT. These instantiations answer different questions and should not be treated as interchangeable.

FWT measures *learning plasticity*: how much the model improves on a task after actually training on it, relative to its state before seeing that task. This is harder to measure than BWT because a standard classifier often has no output units for unseen classes before the task is introduced. Prior work therefore often uses random accuracy as a proxy [Ke et al., 2020, Wołczyk et al., 2021, Ke and

Liu, 2022, Wang et al., 2024] or trains a separate classifier to estimate performance on the next task [Chen et al., 2023]. In this work, we use the following operational form and instantiate it separately for each evaluation interface:

$$FWT = \frac{1}{T-1} \sum_{i=2}^{T} (a_{i,i} - a_{i-1,i}) \tag{2}$$

where T is the last task, $a_{i,i}$ is the score on the i-th task immediately after learning the i-th task, and $a_{i-1,i}$ is the score on the i-th task before training on it. For the trained head, this captures the behavior of the deployed prediction interface. For the probe, it captures how much a newly trained classifier can recover from frozen representations. For the backbone diagnostic, it approximates changes in backbone separability without any trained readout.

We use the term FWT in the operational sense common in continual learning evaluation, but the quantity in Eq. (2) should be read as a task-exposure plasticity gain. It measures how much the model improves on task i after training on task i, relative to the state immediately before seeing that task. This differs from purely zero-shot forward transfer. We keep the term FWT for consistency with IL literature, but throughout the paper we interpret it as an interface-specific plasticity-gain measure.

2.3 Three estimands and a head-only intervention

BWT and operational FWT are formulas over scores $a_{t,i}$, but the score itself is defined by an evaluation interface. Let $m_t = (f_t, g_t)$ denote the model state after training up to task t, where f_t is the backbone and g_t is the sequentially trained prediction head, which we call the trained head. An evaluation interface e specifies which components are frozen, which readout is used or fitted, what auxiliary data are allowed for fitting, and which scoring rule is applied. It induces an interface-specific score

$$a^e_{t,i} = S_e(m_t; D^{\mathrm{tr}}_i, D^{\mathrm{te}}_i),$$

where D^{tr}_i and D^{te}_i denote the task split available to the evaluation protocol. BWT and operational FWT are therefore not properties of m_t alone, but of the pair (m_t, e):

$$\mathrm{BWT}^e = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(a^e_{T,i} - a^e_{i,i} \right),$$

$$\mathrm{FWT}^e_{\mathrm{op}} = \frac{1}{T-1} \sum_{i=2}^{T} \left(a^e_{i,i} - a^e_{i-1,i} \right).$$

Reporting BWT or operational FWT without specifying e is therefore an underspecified measurement. Different interfaces can induce different signs and magnitudes because they instantiate different scoring functionals over the same model trajectory.

The trained head measures deployed behavior, the probe measures recoverability under a newly optimized readout, and the backbone diagnostic measures classifier-free separability. JLT adds a head-only counterfactual intervention for decoder models: it freezes the incrementally trained backbone and asks how much performance can be recovered by re-aligning the LM head.

An interface is defined by what is frozen and which readout is used, not by whether the backbone weights are updated. The backbone diagnostic therefore applies whenever the effective representation $\phi_t(x)$ produced by m_t changes across tasks. This includes prompt-based methods such as L2P [Wang et al., 2022], whose prompts alter the hidden states, and LoRA-based methods such as InfLoRA [Liang and Li, 2024], whose low-rank updates are merged into the backbone. It becomes constant only when the forward pass never changes, as for SEQ* [Zheng et al., 2024] after warm-up or KLDA [Momeni et al., 2025]; the constant value itself indicates zero representation-level plasticity.

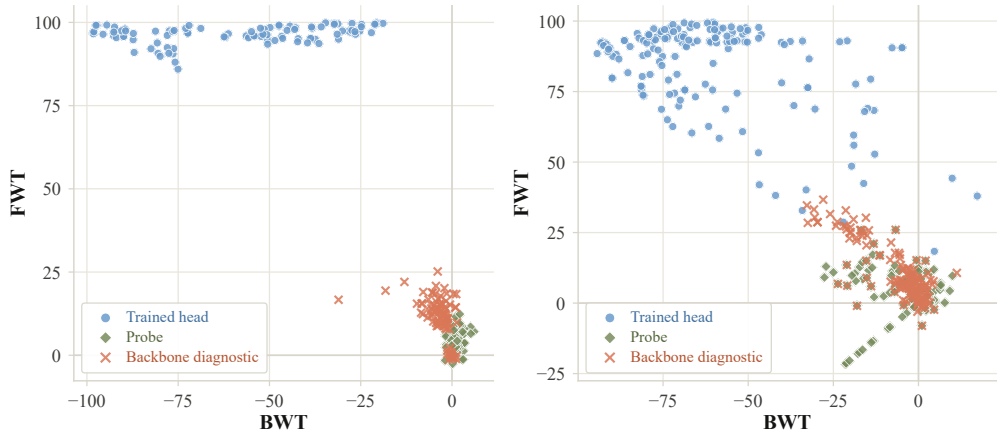
3 Diagnostic interfaces

3.1 Head-level BWT and FWT

The most common evaluation interface is the trained head, the classifier or LM head that is updated sequentially with the backbone. This is the appropriate interface for reporting task accuracy, but its

Table 1: Evaluation interfaces considered in this paper. Each interface instantiates the same BWT/FWT formulas but answers a different measurement question.

Interface	Frozen	Readout	Measures	Does not measure
Trained head	Nothing	Classifier or LM head	Deployed IL performance	Whether the backbone retained information
Probe	Backbone	Newly trained classifier	Recoverable information under a fresh supervised readout	Deployed performance of the IL model
Backbone diagnostic	Nothing	None (clustering, NCM, kNN)	Classifier-free proxy for representation separability	Task accuracy or calibrated prediction
JLT	Backbone	LM head only	Recoverable performance under head re-alignment	Standard CIL performance



(a) Discriminative backbone

(b) Generative backbone

Figure 2: BWT and operational FWT evaluated under three interfaces. Results are aggregated over the benchmark streams in Table 6; the number of incremental tasks varies by dataset.

scores are difficult to attribute to a specific component. Prior work shows that sequentially trained classifiers can become strongly biased toward recently learned classes [Hou et al., 2019, Wu et al., 2019, 2022, Zheng et al., 2024]. As a result, low accuracy on earlier tasks may reflect instability in the head, degradation in the backbone, or both.

Figure 2a illustrates this ambiguity. BWT-head values differ substantially across IL methods, suggesting different degrees of forgetting. However, because the head itself is part of the sequentially updated system, these values do not determine whether the retained representation has disappeared or whether it is no longer read out correctly. Operational FWT-head is even less direct: before a new task is introduced, the head may have no meaningful output for unseen classes.

3.2 Probe-level BWT and FWT

The probe is a linear classifier newly trained on frozen backbone features, as shown in Figure 1. It removes the bias of the trained head and can indicate whether useful information remains accessible in the backbone [Hou et al., 2019, Zhou et al., 2022, Zheng et al., 2024]. However, probe-level evaluation answers a different question from standard task accuracy: it measures what a newly optimized classifier can recover from the representation, not what the incrementally trained model actually predicts.

In Figure 2a, BWT-probe remains close to 0 for most models and IL methods. This suggests that much of the information needed for old tasks is still recoverable. At the same time, a strong probe may compensate for representation shifts or weak separability [Zhou and Srikumar, 2021], thereby masking changes in the backbone itself.

For FWT, the backbone is frozen after each task and the probe is trained over the relevant classes. This makes it possible to evaluate future-task labels, but the measured score depends heavily on the newly trained probe. Thus, the probe measures recoverability under a new readout, but it cannot distinguish an unchanged representation from a shifted one that the probe can still decode.

3.3 Backbone-level BWT and FWT

The backbone has no standalone prediction interface, so backbone-level transfer cannot be observed through accuracy in the same way as head-level transfer. We therefore define the backbone diagnostic as a classifier-free measure of representation separability, instantiated mainly by clustering. As shown in Figure 1, we extract backbone representations after each task, cluster them, align clusters to labels, and use the resulting scores to instantiate BWT-backbone and FWT-backbone. This diagnostic is not intended to replace task accuracy. Instead, it approximates how separable task representations remain independently of any trained readout.

This choice follows a broader line of work that analyzes backbone features to reduce dependence on classifier bias [Liu et al., 2019a, Zhou and Srikumar, 2022, 2021]; related work also uses feature diversity as an auxiliary IL diagnostic [Chen et al., 2023]. Clustering is not a gold-standard measure of representation quality: it is sensitive to feature normalization, the number of clusters, class imbalance, cluster-label matching, the algorithm, and the metric. We therefore treat it as one diagnostic among several and stress-test it across algorithms, metrics, and settings (Section 5.2).

We evaluate clustering diagnostics with five algorithms and three metrics: ACC, ARI, and NMI. Main visualizations use Spectral Clustering, while Table 2 aggregates all clustering algorithms and metrics. We complement clustering with diagnostics that involve no cluster count, initialization, label matching, or trained readout: nearest-class-mean (NCM) and k-nearest-neighbor accuracy, the silhouette score computed with ground-truth labels, and the Fisher discriminant ratio (Appendix E.2). Details are in Appendix B.

4 Experimental setup

4.1 Task & baseline

We evaluate five benchmark groups spanning seven sentence-level datasets, as shown in Table 6. Each benchmark group defines its own incremental stream, with task counts ranging from 5 to 15. For text classification, we use three datasets, AGNews, DBPedia, and Yahoo Answers, as Topic3Datasets [Zhang et al., 2015]. For intent classification, we use CLINC150 [Larson et al., 2019] and Banking77 [Casanueva et al., 2020]. Finally, for relation extraction, we use FewRel [Han et al., 2018] and TACRED [Zhang et al., 2017].

For discriminative backbones, we adopt four representative IL methods: Base, ER [Chaudhry et al., 2019], DER++ [Buzzega et al., 2020], and CLSER [Arani et al., 2022]. For generative backbones, we compare eight baselines: Base, DER++, CLSER, L2KD [Chuang et al., 2020], LAMOL_g, LAMOL_t [Sun et al., 2020], LAMOL_KD [Zheng et al., 2024], and PCLL [Zhao et al., 2022]. We also report recent competitive frozen-backbone methods, such as SEQ* [Zheng et al., 2024] and KLDA [Momeni et al., 2025], in the JLT performance table (Appendix Tables 16 and 17). Because these methods keep the backbone fixed after warm-up or use fixed embeddings, their BWT-backbone and FWT-backbone are constant by construction, so they are not included in the backbone-level correlation analysis. For detailed descriptions of the baselines, please refer to Appendix C.

4.2 Backbone

We evaluate both discriminative backbones (encoder architectures) and generative backbones (decoder architectures). For discriminative backbones, we use BERT-base, BERT-large [Devlin et al., 2019], RoBERTa-base, and RoBERTa-large [Liu et al., 2019b]. For generative backbones, we use Pythia [Biderman et al., 2023] models based on GPT-NeoX [Black et al., 2022] in different sizes (70M, 160M, 410M), as well as Qwen2-0.5B [Yang et al., 2024a], Qwen2.5-0.5B [Yang et al., 2024b], and Qwen3-0.6B [Yang et al., 2025a]. For model-type and model-size analysis, we additionally evaluate the Base method on Pythia-1.4B/2.8B/6.9B, Llama-3.2-1B/3B, Llama-3.1-8B [Dubey et al., 2024], and Qwen3-0.6B/1.7B/4B/8B. For clustering-free diagnostics and parameter-efficient methods,

Table 2: BWT–FWT Pearson correlation at each interface in the CIL scenario. The 150 configurations of the backbone diagnostic combine 10 backbones, five clustering algorithms, and three metrics; they are related rather than independent, so Std summarizes their spread rather than a standard error. Full Pearson and Spearman results are in Appendix E.

Interface	Configurations	Mean r	Std	Range	Sign pattern
Trained head	10 backbones	-0.13	0.38	-0.52 to 0.43	Mixed
Probe	10 backbones	-0.19	0.44	-0.62 to 0.65	Mixed
Backbone diagnostic	150 ($10 \times 5 \times 3$)	-0.74	0.13	-0.93 to -0.43	Negative in all 150

we further evaluate 27 backbone $\times$ dataset $\times$ method units on Llama-3.2-1B and Qwen3-0.6B, including LoRA [Hu et al., 2022] and prompt-tuning variants (Appendix E.2). More details on training and backbone evaluation are in Appendix A.1.

5 Results

5.1 The same BWT/FWT metric yields different conclusions

The three interfaces do not merely differ in magnitude; they support different qualitative conclusions. In discriminative backbones, the per-method plots in Appendix Figure 7 show that BWT-head suggests severe forgetting, especially for Base, because the trained head is strongly biased toward recently learned classes. BWT-probe remains close to 0, suggesting that a newly trained probe can recover much of the old-task information. BWT-backbone lies between these two views: the representation itself changes, but not as drastically as head-level evaluation implies.

The operational FWT results show the same interface dependence. FWT-head values can exceed 80, largely reflecting the introduction of task-specific output classes rather than pure transfer. FWT-probe remains small (below 5 on average) because the probe can compensate for representation differences. By contrast, FWT-backbone ranges over a meaningful intermediate scale, indicating measurable changes in representation separability before and after task training.

The generative-backbone results in Figures 2b and 3 mirror this pattern. BWT-head again indicates severe forgetting (e.g., -86.7 on average for Base), while the backbone diagnostic shows that decoder representations retain substantially more structure than the LM head can read out directly. This suggests that the backbone–head gap is not limited to encoder classifiers.

Finding 1. The trained head suggests catastrophic forgetting, the probe suggests strong recoverability, and the backbone diagnostic indicates partial representation drift. Forgetting is therefore not a single model-level number; it depends on the interface through which it is measured.

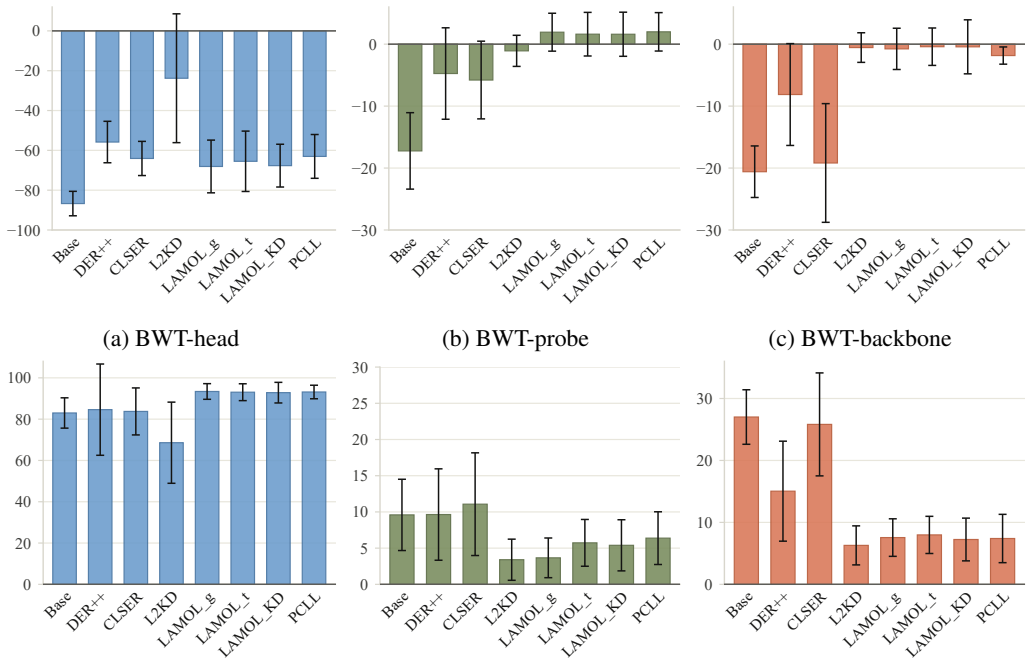
5.2 Backbone trends are consistent across diagnostic choices

BWT and FWT share the intermediate term $a_{i,i}$:

$$BWT_i = a_{T,i} - a_{i,i}, \qquad FWT_i = a_{i,i} - a_{i-1,i}.$$

This coupling can make a negative correlation partly expected by construction, so the sign alone is not informative. What matters is how consistently and how strongly the relation holds across interfaces. As Table 2 shows, the trained head and probe interfaces have mixed signs and weak aggregate correlations, whereas the backbone diagnostic is negative in every clustering algorithm, metric, and backbone combination (mean $r = -0.74$). The 150 configurations are related rather than independent samples, so we use them to assess robustness, not to estimate a sampling distribution.

Because clustering scores depend on normalization, the number of clusters, initialization, and label matching, we also evaluate clustering-free diagnostics on 27 additional backbone $\times$ dataset $\times$ method units (Appendix Table 14). NCM stays strongly negative in every subset (-0.56 to -0.92), whereas the trained-head correlation changes sign (-1.00 to $+0.33$). Over all 27 units, every clustering-free diagnostic is negative (NCM -0.87, silhouette -0.84, kNN-10 -0.80, kNN-1 -0.62, Fisher ratio -0.50), and the clustering-based correlation remains negative under all 25 settings of normalization, cluster count, seed, label matching, class balance, sample size, and algorithm, for ACC, ARI, and



(a) BWT-head (b) BWT-probe (c) BWT-backbone

(d) FWT-head (e) FWT-probe (f) FWT-backbone

Figure 3: Average and standard deviation of head-level, probe-level, and backbone-level BWT/FWT for each IL method in six generative backbones, aggregated over the benchmark streams in Table 6.

NMI alike (Appendix B.3). Clustering-based and clustering-free diagnostics therefore agree, and the representation-level relation does not hinge on clustering. Full Pearson and Spearman correlations for the main study are reported in Tables 12 and 13.

Finding 2. The representation-level stability–plasticity pattern is consistent across diagnostic choices, while the trained-head and probe interfaces remain unstable in sign and magnitude.

5.3 Stability–plasticity trade-off in the backbone

Strong IL results can suggest that the stability–plasticity dilemma has largely been addressed, but many of these gains come from keeping the backbone close to its earlier state while adjusting the classifier or LM head. Our backbone diagnostic suggests that the trade-off persists at the representation level: BWT-backbone and FWT-backbone are negatively correlated across backbones and diagnostics (Table 2), so configurations that retain more old-task structure tend to gain less plasticity on new tasks. Better head-level performance therefore does not by itself show that the backbone keeps learning new tasks effectively.

Parameter-efficient methods. The diagnostic also applies to parameter-efficient fine-tuning (PEFT), because LoRA updates and prompts change the effective representation $\phi_t(x)$ (Section 2.3). On Llama-3.2-1B, LoRA still shows severe forgetting at the trained head (BWT-head -76.5 to -89.0) and prompt tuning keeps head accuracy low throughout (A_T 4.6–10.2), while the representation-level changes are much smaller (Appendix Table 15). On Banking77, LoRA is more stable than full fine-tuning with Base at the backbone (BWT-backbone -1.91 vs. -3.71) but has less than half of its plasticity (FWT-backbone 2.70 vs. 6.06, and 16.89 for CLSER), although all three methods show BWT-head below -82. Parameter isolation thus buys stability at the cost of representation-level plasticity, a trade-off that is invisible at the trained head.

Finding 3. The backbone diagnostic does not support the view that current IL methods have resolved the stability–plasticity dilemma at the representation level.

5.4 Secondary observation: scaling improves retention more than plasticity

In the Base setting, increasing model size within Pythia, Llama, and Qwen improves BWT-backbone more consistently than FWT-backbone (Appendix Figure 10), suggesting that scale helps retention more than the immediate plasticity gain measured by our operational FWT.

6 A Head-Only Intervention Estimates Recoverable Forgetting

The three-interface analysis suggests that poor standard accuracy may not imply that the backbone has erased the task. To test this interpretation, after an IL method has trained the model sequentially, we freeze the backbone and re-optimize only the LM head with a label-token cross-entropy objective, without adding adapters, external encoders, or independent classifiers (Appendix F; full results in Appendix Tables 16–17).

Table 3: Aggregate JLT decomposition over three decoder backbones and five benchmark groups (task-adaptive protocol; Appendix Tables 16–17). $G_{\mathrm{JLT}} = A_T^{\mathrm{JLT}} - A_T^{\mathrm{standard}}$, Residual $= A_T^{\mathrm{joint}} - A_T^{\mathrm{JLT}}$ with $A_T^{\mathrm{joint}} = 96.38$, and Recovery $= G_{\mathrm{JLT}}/(A_T^{\mathrm{joint}} - A_T^{\mathrm{standard}})$.

	A_T					BWT		FWT	
IL method	Standard	JLT	G_{JLT}	Residual	Recovery	Standard	JLT	Standard	JLT
Base	11.07	74.73	+63.65	21.65	74.6%	−87.79	−19.12	86.55	70.91
L2KD	40.72	95.56	+54.84	0.82	98.5%	−42.75	−1.76	74.77	75.86
LAMOL_KD	40.63	94.14	+53.50	2.25	96.0%	−62.39	−2.68	93.19	74.47
PCLL	41.63	94.51	+52.88	1.87	96.6%	−61.48	−1.89	90.48	76.21

JLT is not meant to compete with standard IL protocols but to measure how much performance would be recoverable if only the readout were re-aligned. We therefore report the JLT gap,

$$G_{\mathrm{JLT}} = A_T^{\mathrm{JLT}} - A_T^{\mathrm{standard}},$$

as the primary quantity. A large gap indicates that the loss in standard accuracy is not fully explained by representation erasure; a small gap combined with low standard accuracy would instead indicate that the backbone itself has lost the relevant information.

The aggregate results in Table 3 show that all four IL methods have large recoverable gaps. Although standard evaluation indicates severe forgetting, freezing the backbone and re-aligning only the LM head recovers much of the performance. For example, L2KD averages 40.72 under standard evaluation but 95.56 under the diagnostic head-realignment protocol, reducing average BWT from -42.75 to -1.76. Over the same models and benchmarks, joint fine-tuning averages 96.38 in A_T, while L2KD+JLT averages 95.56, motivating a diagnostic decomposition of the gap to the joint-training reference.

Let A_T^{joint} denote the joint-training reference. The gap from standard IL evaluation to the joint reference can be decomposed as

$$A_T^{\mathrm{joint}} - A_T^{\mathrm{standard}} = \underbrace{A_T^{\mathrm{JLT}} - A_T^{\mathrm{standard}}}_{\text{readout-recoverable gap}} + \underbrace{A_T^{\mathrm{joint}} - A_T^{\mathrm{JLT}}}_{\text{residual gap}}.$$

We use this decomposition only diagnostically. The first term estimates the portion of apparent forgetting that can be removed by re-aligning the LM head on a fixed incrementally trained backbone; the second term is the residual gap that head re-alignment alone does not remove. This decomposition shows that recoverability is method-dependent (Table 3). L2KD, LAMOL_KD, and PCLL recover 96–99% of the gap to the joint reference, whereas Base recovers only 74.6% (74.0–75.2% across the three backbones). The residual is also stream-specific: for Base it is close to zero on TACRED (-0.3 to 1.2) but about 44 points on FewRel for every backbone, while for L2KD it is at most 4.34 points on every stream (Appendix Table 18). Neither pattern is visible from standard accuracy or from G_{JLT} alone.

Table 4 turns the two terms into a two-axis reading. A method with poor standard accuracy but a large G_{JLT} and a small residual, such as L2KD, is readout-limited: its representation remains usable, and improving it requires changes to the head rather than to the backbone. A large residual instead calls for backbone-level intervention, as for Base on FewRel.

Table 4: Two-axis reading of the JLT decomposition. Examples use Table 3 and Appendix Table 18.

	Small residual (representation preserved)	**Large residual** (representation lost)
Large G_{JLT}	*Readout-limited*: realign or preserve the head (e.g., L2KD, residual 0.82)	*Mixed*: intervene on both head and backbone (e.g., Base, residual 21.65; FewRel ≈44)
Small G_{JLT}	*Already aligned*	*Representation-limited*: replay, regularization, or another backbone-level intervention

Effect of head re-alignment on FWT. Under task-adaptive JLT, both $a_{i,i}$ and $a_{i-1,i}$ are measured with re-aligned heads, so JLT-FWT excludes the gain that comes from creating an output interface for previously unseen labels. On the relation-extraction streams (TACRED and FewRel), FWT drops by 29.5–40.4 points on average for every method; for Pythia-410M with L2KD on TACRED, it falls from 94.39 to 4.92. On the intent and topic streams, the change is small for Base, LAMOL_KD, and PCLL (-4.3 to -1.1), and L2KD even gains 21.5 points, mainly on Topic3, where its standard FWT is only about 40 (Appendix Table 18). Most of the head-level FWT on relation extraction therefore reflects interface construction, whereas on intent and topic streams it largely reflects plasticity that persists after the readout is aligned.

These results should not be read as evidence that head-only tuning is a deployable replacement for continual learning. The task-adaptive JLT variant adapts the head before evaluating each task and therefore estimates recoverable performance rather than standard CIL accuracy. A deployable variant, post-hoc JLT, re-aligns the head once per checkpoint under the same rehearsal assumption as ER, using a small buffer drawn from the tasks seen so far with roughly one example per class, and never accesses the full training split of an evaluated task. On Llama-3.2-1B, it improves standard A_T by 23.8 points and BWT by 52.6 points on average over four dataset–method pairs, while FWT falls from 94.3 to 70.7 as the interface-construction component is removed (Appendix F.2).

Finding 4. With the incrementally trained backbone fixed, re-aligning only the LM head recovers 96–99% of the gap to joint training for replay- and distillation-based methods but only about three quarters for Base, whose unrecovered residual concentrates on specific streams such as FewRel.

Threats to interpretation. Our backbone diagnostic is not task accuracy; it is a classifier-free separability proxy. The clustering configurations are not independent statistical samples, but robustness stress tests over related diagnostic choices. Because BWT and operational FWT share $a_{i,i}$, we interpret their relation comparatively across interfaces rather than as proof of a universal trade-off. Finally, task-adaptive JLT is oracle-style: it measures recoverability under head re-alignment, not deployable CIL performance.

7 Conclusion

Our central message is that forgetting in incremental learning is not a single model-level quantity. It is an interface-dependent measurement. The trained head measures deployed behavior, the probe measures recoverability under a new readout, and the backbone diagnostic measures classifier-free separability. These views can disagree sharply, and treating them as interchangeable can lead to misleading conclusions about stability and plasticity. The JLT intervention shows why this distinction matters: when the incrementally trained backbone is fixed and only the LM head is re-aligned, much of the apparent forgetting is recoverable. This does not mean that standard IL is solved, nor that head-only tuning is a deployable replacement for continual learning. It means that many failures attributed to catastrophic forgetting are better understood as a combination of partial representation drift and readout mismatch. We therefore recommend that future incremental learning work report not only BWT, FWT, and average accuracy, but also the interface through which these quantities are measured.

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References

Wickliffe C. Abraham and Anthony Robins. Memory retention – the synaptic stability versus plasticity dilemma. *Trends in Neurosciences*, 28(2):73–78, 2005. doi: 10.1016/j.tins.2004.12.003. URL https://doi.org/10.1016/j.tins.2004.12.003.

Elahe Arani, Fahad Sarfraz, and Bahram Zonooz. Learning fast, learning slow: A general continual learning method based on complementary learning system. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=uxxFrDwrE7Y.

Vladimir Araujo, Julio Hurtado, Alvaro Soto, and Marie-Francine Moens. Entropy-based stability-plasticity for lifelong learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 3721–3728, 2022. URL https://openaccess.thecvf.com/content/CVPR2022W/CLVision/html/Araujo_Entropy-Based_Stability-Plasticity_for_Lifelong_Learning_CVPRW_2022_paper.html.

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usvsn Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. Pythia: A suite for analyzing large language models across training and scaling. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR, 2023. URL https://proceedings.mlr.press/v202/biderman23a.html.

Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, Usvsn Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In Angela Fan, Suzana Ilic, Thomas Wolf, and Matthias Gallé, editors, *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*, pages 95–136, virtual+Dublin, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.bigscience-1.9. URL https://aclanthology.org/2022.bigscience-1.9/.

Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: a strong, simple baseline. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 15920–15930. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/b704ea2c39778f07c617f6b7ce480e9e-Paper.pdf.

Iñigo Casanueva, Tadas Temčinas, Daniela Gerz, Matthew Henderson, and Ivan Vulić. Efficient intent detection with dual sentence encoders. In Tsung-Hsien Wen, Asli Celikyilmaz, Zhou Yu, Alexandros Papangelis, Mihail Eric, Anuj Kumar, Iñigo Casanueva, and Rushin Shah, editors, *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, pages 38–45, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.nlp4convai-1.5. URL https://aclanthology.org/2020.nlp4convai-1.5/.

Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, Puneet K. Dokania, Philip H. S. Torr, and Marc'Aurelio Ranzato. On tiny episodic memories in continual learning. *arXiv preprint arXiv:1902.10486*, 2019. URL https://arxiv.org/abs/1902.10486.

Jiefeng Chen, Timothy Nguyen, Dilan Gorur, and Arslan Chaudhry. Is forgetting less a good inductive bias for forward transfer? In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=dL35lx-mTEs.

Yung-Sung Chuang, Shang-Yu Su, and Yun-Nung Chen. Lifelong language knowledge distillation. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2914–2924, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.233. URL https://aclanthology.org/2020.emnlp-main.233/.

MohammadReza Davari, Nader Asadi, Sudhir Mudur, Rahaf Aljundi, and Eugene Belilovsky. Probing representation forgetting in supervised and unsupervised continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16712–16721, June 2022.

Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1): 1–22, 1977. doi: 10.1111/j.2517-6161.1977.tb01600.x.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL https://aclanthology.org/N19-1423/.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Sayna Ebrahimi, Mohamed Elhoseiny, Trevor Darrell, and Marcus Rohrbach. Uncertainty-guided continual learning in Bayesian neural networks – extended abstract. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2019. URL https://openaccess.thecvf.com/content_CVPRW_2019/html/Uncertainty_and_Robustness_in_Deep_Visual_Learning/Ebrahimi_Uncertainty-Guided_Continual_Learning_in_Bayesian_Neural_Networks_-_Extended_Abstract_CVPRW_2019_paper.html.

Sayna Ebrahimi, Franziska Meier, Roberto Calandra, Trevor Darrell, and Marcus Rohrbach. Adversarial continual learning. In *Computer Vision – ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XI*, volume 12356 of *Lecture Notes in Computer Science*, pages 386–402. Springer, 2020. doi: 10.1007/978-3-030-58621-8_23. URL https://doi.org/10.1007/978-3-030-58621-8_23.

Kawin Ethayarajh. How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65, Hong Kong, China, 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1006. URL https://aclanthology.org/D19-1006/.

Robert M. French. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3 (4):128–135, 1999. doi: 10.1016/S1364-6613(99)01294-2. URL https://doi.org/10.1016/S1364-6613(99)01294-2.

Xu Han, Hao Zhu, Pengfei Yu, Ziyun Wang, Yuan Yao, Zhiyuan Liu, and Maosong Sun. FewRel: A large-scale supervised few-shot relation classification dataset with state-of-the-art evaluation. In Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii, editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 4803–4809, Brussels, Belgium, October-November 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1514. URL https://aclanthology.org/D18-1514/.

Zeyu Han, Chao Gao, Jinyang Liu, Jeff Zhang, and Sai Qian Zhang. Parameter-efficient fine-tuning for large models: A comprehensive survey. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=lIsCS8b6zj.

Saihui Hou, Xinyu Pan, Chen Change Loy, Zilei Wang, and Dahua Lin. Learning a unified classifier incrementally via rebalancing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 831–839, June 2019. doi: 10.1109/CVPR.2019.00092.

Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2790–2799. PMLR, 2019. URL https://proceedings.mlr.press/v97/houlsby19a.html.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Wenpeng Hu, Zhou Lin, Bing Liu, Chongyang Tao, Zhengwei Tao, Jinwen Ma, Dongyan Zhao, and Rui Yan. Overcoming catastrophic forgetting for continual learning via model adaptation. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=ryGvcoA5YX.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.

Stephen C. Johnson. Hierarchical clustering schemes. *Psychometrika*, 32(3):241–254, 1967. doi: 10.1007/BF02289588. URL https://doi.org/10.1007/BF02289588.

Prakhar Kaushik, Alex Gain, Adam Kortylewski, and Alan Yuille. Understanding catastrophic forgetting and remembering in continual learning with optimal relevance mapping. *arXiv preprint arXiv:2102.11343*, 2021. URL https://arxiv.org/abs/2102.11343.

Zixuan Ke and Bing Liu. Continual learning of natural language processing tasks: A survey. *arXiv preprint arXiv:2211.12701*, 2022. URL https://arxiv.org/abs/2211.12701.

Zixuan Ke, Bing Liu, and Xingchang Huang. Continual learning of a mixed sequence of similar and dissimilar tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 18493–18504. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/d7488039246a405baf6a7cbc3613a56f-Abstract.html.

Ronald Kemker and Christopher Kanan. FearNet: Brain-inspired model for incremental learning. In *International Conference on Learning Representations*, 2018. URL https://openreview.net/forum?id=SJ1Xmf-Rb.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017. doi: 10.1073/pnas.1611835114.

Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. An evaluation dataset for intent classification and out-of-scope prediction. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1311–1316, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1131. URL https://aclanthology.org/D19-1131/.

Xilai Li, Yingbo Zhou, Tianfu Wu, Richard Socher, and Caiming Xiong. Learn to grow: A continual structure learning framework for overcoming catastrophic forgetting. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3925–3934. PMLR, 2019. URL https://proceedings.mlr.press/v97/li19m.html.

Yan-Shuo Liang and Wu-Jun Li. InfLoRA: Interference-free low-rank adaptation for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23638–23647, June 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Liang_InfLoRA_Interference-Free_Low-Rank_Adaptation_for_Continual_Learning_CVPR_2024_paper.html.

Tianlin Liu, Lyle Ungar, and João Sedoc. Continual learning for sentence representations using conceptors. In Jill Burstein, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3274–3279, Minneapolis, Minnesota, June 2019a. Association for Computational Linguistics. doi: 10.18653/v1/N19-1331. URL https://aclanthology.org/N19-1331/.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *CoRR*, abs/1907.11692, 2019b. URL http://arxiv.org/abs/1907.11692.

David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/f87522788a2be2d171666752f97ddebb-Abstract.html.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

J. MacQueen. Some methods for classification and analysis of multivariate observations. In Lucien M. Le Cam and Jerzy Neyman, editors, *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Statistics*, pages 281–297, Berkeley, CA, 1967. University of California Press. URL https://projecteuclid.org/euclid.bsmsp/1200512992.

Martial Mermillod, Aurélia Bugaiska, and Patrick Bonin. The stability-plasticity dilemma: investigating the continuum from catastrophic forgetting to age-limited learning effects. *Frontiers in Psychology*, 4:504, 2013. doi: 10.3389/fpsyg.2013.00504. URL https://doi.org/10.3389/fpsyg.2013.00504.

Saleh Momeni, Sahisnu Mazumder, and Bing Liu. Continual learning using a kernel-based method over foundation models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 19528–19536, 2025. doi: 10.1609/aaai.v39i18.34150. URL https://doi.org/10.1609/aaai.v39i18.34150.

Andrew Y. Ng, Michael I. Jordan, and Yair Weiss. On spectral clustering: Analysis and an algorithm. In T. Dietterich, S. Becker, and Z. Ghahramani, editors, *Advances in Neural Information Processing Systems*, volume 14, pages 849–856. MIT Press, 2001. URL https://proceedings.neurips.cc/paper_files/paper/2001/file/801272ee79cfde7fa5960571fee36b9b-Paper.pdf.

German I. Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. Continual lifelong learning with neural networks: A review. *Neural Networks*, 113:54–71, 2019. doi: 10.1016/j.neunet.2019.01.012.

Robi Polikar, Lalita Udpa, Satish S. Udpa, and Vasant Honavar. Learn++: an incremental learning algorithm for supervised neural networks. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 31(4):497–508, 2001. doi: 10.1109/5326.983933. URL https://doi.org/10.1109/5326.983933.

Fan-Keng Sun, Cheng-Hao Ho, and Hung-Yi Lee. LAMOL: LAnguage MOdeling for lifelong language learning. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=Skgxcn4YDS.

Mingxu Tao, Yansong Feng, and Dongyan Zhao. Can BERT refrain from forgetting on sequential tasks? A probing study. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=UazgYBMS9-W.

Gido M. van de Ven, Nicholas Soures, and Dhireesha Kudithipudi. Continual learning and catastrophic forgetting. In John H. Wixted, editor, *Learning and Memory: A Comprehensive Reference (Third Edition)*, volume 1, pages 153–168. Academic Press, Oxford, 2025. doi: 10.1016/B978-0-443-15754-7.00073-0. URL https://doi.org/10.1016/B978-0-443-15754-7.00073-0.

Nguyen Xuan Vinh, Julien Epps, and James Bailey. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11(95):2837–2854, 2010. URL http://jmlr.org/papers/v11/vinh10a.html.

Liyuan Wang, Xingxing Zhang, Hang Su, and Jun Zhu. A comprehensive survey of continual learning: Theory, method and application. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(8):5362–5383, 2024. doi: 10.1109/TPAMI.2024.3367329. URL https://doi.org/10.1109/TPAMI.2024.3367329.

Zhicheng Wang, Yufang Liu, Tao Ji, Xiaoling Wang, Yuanbin Wu, Congcong Jiang, Ye Chao, Zhencong Han, Ling Wang, Xu Shao, and Wenqiu Zeng. Rehearsal-free continual language learning via efficient parameter isolation. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10933–10946, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.612. URL https://aclanthology.org/2023.acl-long.612/.

Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. Learning to prompt for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 139–149, June 2022.

Maciej Wołczyk, Michał Zając, Razvan Pascanu, Łukasz Kuciński, and Piotr Miłoś. Continual world: A robotic benchmark for continual reinforcement learning. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 28496–28510. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/ef8446f35513a8d6aa2308357a268a7e-Paper.pdf.

Guile Wu, Shaogang Gong, and Pan Li. Striking a balance between stability and plasticity for class-incremental learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1124–1133, October 2021.

Tongtong Wu, Massimo Caccia, Zhuang Li, Yuan-Fang Li, Guilin Qi, and Gholamreza Haffari. Pretrained language model in continual learning: A comparative study. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=figzpGMrdD.

Yue Wu, Yinpeng Chen, Lijuan Wang, Yuancheng Ye, Zicheng Liu, Yandong Guo, and Yun Fu. Large scale incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 374–382, 2019. doi: 10.1109/CVPR.2019.00046. URL https://openaccess.thecvf.com/content_CVPR_2019/html/Wu_Large_Scale_Incremental_Learning_CVPR_2019_paper.html.

Junyuan Xie, Ross Girshick, and Ali Farhadi. Unsupervised deep embedding for clustering analysis. In Maria Florina Balcan and Kilian Q. Weinberger, editors, *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 478–487. PMLR, 2016. URL https://proceedings.mlr.press/v48/xieb16.html.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhang Zhu, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Xuejing Liu, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, Zhifang Guo, and Zhihao Fan. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024a. doi: 10.48550/arXiv.2407.10671. URL https://arxiv.org/abs/2407.10671.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024b. doi: 10.48550/arXiv.2412.15115. URL https://arxiv.org/abs/2412.15115.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a. doi: 10.48550/arXiv.2505.09388. URL https://arxiv.org/abs/2505.09388.

Yutao Yang, Jie Zhou, Xuanwen Ding, Tianyu Huai, Shunyu Liu, Qin Chen, Yuan Xie, and Liang He. Recent advances of foundation language models-based continual learning: A survey. *ACM Computing Surveys*, 57(5):1–38, 2025b. doi: 10.1145/3705725.

Tiantian Zhang, Li Zhong, and Bo Yuan. A critical note on the evaluation of clustering algorithms. *arXiv preprint arXiv:1908.03782*, 2019. URL https://arxiv.org/abs/1908.03782.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 28, pages 649–657. Curran Associates, Inc., 2015. URL https://proceedings.neurips.cc/paper_files/paper/2015/file/250cf8b51c773f3f8dc8b4be867a9a02-Paper.pdf.

Yuhao Zhang, Victor Zhong, Danqi Chen, Gabor Angeli, and Christopher D. Manning. Position-aware attention and supervised data improve slot filling. In Martha Palmer, Rebecca Hwa, and Sebastian Riedel, editors, *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 35–45, Copenhagen, Denmark, 2017. Association for Computational Linguistics. doi: 10.18653/v1/D17-1004. URL https://aclanthology.org/D17-1004/.

Yingxiu Zhao, Yinhe Zheng, Zhiliang Tian, Chang Gao, Jian Sun, and Nevin L. Zhang. Prompt conditioned VAE: Enhancing generative replay for lifelong learning in task-oriented dialogue. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11153–11169, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.766. URL https://aclanthology.org/2022.emnlp-main.766/.

Junhao Zheng, Shengjie Qiu, and Qianli Ma. Learn or recall? revisiting incremental learning with pre-trained language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14848–14877, Bangkok, Thailand, August 2024. Association for Computational

Linguistics. doi: 10.18653/v1/2024.acl-long.794. URL https://aclanthology.org/2024.acl-long.794/.

Junhao Zheng, Xidi Cai, Shengjie Qiu, and Qianli Ma. Spurious forgetting in continual learning of language models. In *International Conference on Learning Representations*, 2025a. URL https://openreview.net/forum?id=ScI7IlKGdI.

Junhao Zheng, Shengjie Qiu, Chengming Shi, and Qianli Ma. Towards lifelong learning of large language models: A survey. *ACM Computing Surveys*, 57(8):1–35, 2025b. doi: 10.1145/3716629.

Da-Wei Zhou, Fu-Yun Wang, Han-Jia Ye, Liang Ma, Shiliang Pu, and De-Chuan Zhan. Forward compatible few-shot class-incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9046–9056, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Zhou_Forward_Compatible_Few-Shot_Class-Incremental_Learning_CVPR_2022_paper.html.

Yichu Zhou and Vivek Srikumar. DirectProbe: Studying representations without classifiers. In Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou, editors, *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5070–5083, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.401. URL https://aclanthology.org/2021.naacl-main.401/.

Yichu Zhou and Vivek Srikumar. A closer look at how fine-tuning changes BERT. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1046–1061, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.75. URL https://aclanthology.org/2022.acl-long.75/.

Limitations

We use classifier-free diagnostics to approximate changes in backbone separability, but they are not a complete substitute for head-level task accuracy. Clustering scores in particular depend on the algorithm, number of clusters, feature normalization, class balance, cluster-label matching, and evaluation metric, which is why we also report clustering-free diagnostics and a sensitivity analysis. Our goal is therefore not to propose a perfect new evaluation metric, but to provide an auxiliary view of how the backbone representation changes during IL.

JLT also has an important limitation: the task-adaptive variant performs head-only adaptation before evaluating each task. This should be interpreted as an oracle-style diagnostic of recoverable information in the backbone, not as a deployable CIL protocol that has access to each test task's training data at evaluation time. The reported gains are useful for quantifying head–backbone mismatch, but they should not be read as direct standard IL performance.

Sequential fine-tuning across many tasks is computationally expensive, which limited our experiments with larger generative models. We attempted to use Qwen3-32B [Yang et al., 2025a], Llama-3-70B [Dubey et al., 2024], and related models, but could not run them within our compute budget or apply every IL method under the same conditions. Models larger than 8B typically require PEFT [Houlsby et al., 2019, Hu et al., 2022, Han et al., 2024]; in that setting, several IL methods (L2KD, LAMOL, LAMOL_KD) did not work reliably. Our experiments are therefore limited to models below 10B parameters. Proposing a new IL method that optimizes the backbone-level stability–plasticity trade-off remains future work.

Broader impacts

This work does not release a new pretrained model or dataset. Its positive impact is to make continual learning evaluation more transparent by separating deployed performance from recoverable representation information. By re-examining BWT and FWT through the trained head, a probe, and the backbone diagnostic, we show that transfer metrics can lead to different conclusions depending on where they are measured, which supports more careful and reproducible evaluation.

A possible negative impact is that diagnostic recoverability could be mistaken for deployable performance, leading to overoptimistic claims about model updating. We mitigate this by explicitly separating standard IL results from diagnostic JLT recoverability and by recommending that future work report the evaluation interface used for each metric.

A Experimental details

A.1 Backbone details

For the discriminative backbone, learning and evaluation are performed using the [CLS] token features of the last hidden states [Ethayarajh, 2019]. The classifier for each task is a linear layer whose input dimension is the hidden size of the backbone and whose output dimension is the number of classes in that task. In TIL, each task is trained and evaluated with its own classifier; in CIL, the logits of all task classifiers are concatenated. To use the generative backbone in sentence-level tasks, we use two types of prompts depending on the task type. For text and intent classification, we use the following prompt:

```
"Input sentence: {text}\n The Label: {label}{eos token}"
```

For relation extraction, we use the following prompt:

```
"Input sentence: {text}\n The relationship between {head entity} and {tail
entity} is {label}{eos token}"
```

With these prompts, we optimize the causal language modeling loss on the `{label}{eos token}` span [Zheng et al., 2024]. Depending on the IL method, additional objectives such as cross-entropy, KL-divergence, or MSE distillation losses are used.

Table 5: Estimated compute resources for the reported experiments. Each run uses a single GPU, and eight GPUs were used in parallel. Runtimes are approximate ranges measured on our local GPUs. The number of runs counts benchmark-stream/model/method/seed configurations used for the reported results.

Experiment group	GPU	Runtime per run	# Runs	GPU-hours
Discriminative CIL	8× RTX A5000 (24GB)	0.5–3 h	≈24	≈12–72
Generative CIL, small models	8× RTX A5000 (24GB)	1–4 h	≈72	≈72–288
JLT diagnostic	8× RTX A5000 (24GB)	0.2–1 h	≈18	≈3.6–18
Model-size analysis	8× RTX A5000 (24GB)	2–8 h	≈24	≈48–192

Table 6: Statistics of the incremental streams built from the sentence-level datasets used in our experiments. **# Tasks** is the number of incremental tasks, **# CIL classes** is the number of test-set classes in the CIL scenario, and **# TIL classes** is the number of test-set classes evaluated for each task in the TIL scenario.

Task	Dataset	# Classes	# Tasks	# CIL classes	# TIL classes	# Training instances	# Test instances
Text classification	Topic3Datasets	25	5	25	5	75,000	46,000
Intent classification	CLINC150	150	15	150	10	15,000	4,500
	Banking77	77	7	77	11	7,191	2,800
Relation extraction	FewRel	80	8	80	10	33,600	11,200
	TACRED	40	8	40	5	5,909	1,259

A.2 Implementation details

For Topic3Datasets, we use a maximum sequence length of 256 and 3 epochs per incremental task; for FewRel and TACRED, 128 and 5 epochs; and for CLINC150 and Banking77, 64 and 5 epochs. We used a learning rate of 1×10^{-5} for each backbone and 1×10^{-3} for the classifier with the AdamW optimizer [Loshchilov and Hutter, 2019]. Probes are linear classifiers trained for 20 epochs on frozen backbone features.

All experiments were run on eight NVIDIA RTX A5000 GPUs (24GB each). Unless otherwise noted, each experiment was run with three random seeds, reported means average over the seeds, and error bars in figures denote one standard deviation across seeds. Correlation coefficients are computed over the corresponding dataset–method units for each evaluation interface. The compute estimates in Table 5 are approximate and are intended to support reproducibility planning rather than exact accounting. For the remaining hyperparameters of each IL method, we follow the settings recommended in the original papers.

For assets whose public dataset cards do not specify a single explicit license, we use the datasets only under their commonly distributed research benchmark setting and cite the original sources. We do not redistribute the raw datasets.

A.3 Evaluation metrics

A.3.1 Average accuracy

The most common way to evaluate IL is to measure performance on all tasks after the final task has been learned, and Average Accuracy is reported in most studies. When T tasks are evaluated, however, $T-1$ of them are previous tasks, so this metric rewards retention much more than learning the current task. Average Accuracy after task t is defined as

$$A_t = \frac{1}{t} \sum_{i=1}^{t} a_{t,i} \tag{3}$$

where $a_{t,i}$ is the accuracy on the test set of task i after the model has been trained incrementally on tasks 1 to t.

Table 7: Existing datasets and pretrained model assets used in this work. We cite the original papers in the main text and summarize the public license or access terms reported by the corresponding source or model card.

Asset	Type	Version / source	License or access terms
Topic3Datasets (AGNews, DBPedia, Yahoo)	Dataset	Zhang et al. topic-classification datasets; DBpedia source data from DBpedia releases	AGNews/Yahoo: research-use terms from the original sources; DBpedia: CC-BY-SA/GFDL terms for DBpedia releases
CLINC150	Dataset	UCI Machine Learning Repository, `https://archive.ics.uci.edu/dataset/570/clinc150`	CC BY 4.0
Banking77	Dataset	PolyAI-LDN dataset card, `https://huggingface.co/datasets/PolyAI/banking77`	CC BY 4.0
FewRel	Dataset	THUNLP FewRel release, `https://huggingface.co/datasets/thunlp/few_rel`	MIT License
TACRED	Dataset	Stanford TACRED / LDC2018T24, `https://catalog.ldc.upenn.edu/LDC2018T24`	LDC User Agreement
BERT	Model	Google BERT model card, e.g., `https://huggingface.co/google-bert/bert-base-uncased`	Apache License 2.0
RoBERTa	Model	FacebookAI RoBERTa model card, e.g., `https://huggingface.co/FacebookAI/roberta-base`	MIT License
Pythia / GPT-NeoX	Model	EleutherAI Pythia model cards, e.g., `https://huggingface.co/EleutherAI/pythia-410m-v0`	Apache License 2.0
Qwen2, Qwen2.5, Qwen3	Model	Qwen model cards, e.g., `https://huggingface.co/Qwen/Qwen3-0.6B`	Apache License 2.0
Llama 3.1 / 3.2	Model	Meta Llama model cards, e.g., `https://huggingface.co/meta-llama/Llama-3.2-3B`	Llama Community License and Acceptable Use Policy

A.3.2 Average incremental accuracy

Average Incremental Accuracy averages A_t over all steps $t = 1, \ldots, T$. It avoids judging a method by A_T alone, but it usually follows the same trend as Average Accuracy because previous-task accuracy still dominates the average.

$$\bar{A} = \frac{1}{T} \sum_{t=1}^{T} A_t \tag{4}$$

B Clustering algorithm details

To evaluate the features extracted from the backbone, we employ five representative clustering algorithms: K-means [MacQueen, 1967], Gaussian Mixture Model [Dempster et al., 1977], Spectral Clustering [Ng et al., 2001], Agglomerative Clustering [Johnson, 1967], and Deep Clustering [Xie et al., 2016]. While all algorithms are evaluated, the main visualizations use Spectral Clustering and the aggregate summary in Table 2 uses all algorithms and metrics.

B.1 Clustering evaluation metrics

ACC (Accuracy)

We build a contingency matrix between predicted clusters and ground-truth labels, find the one-to-one cluster-to-label mapping that maximizes agreement with the Hungarian algorithm, and report accuracy under this mapping.

ARI (Adjusted Rand Index)

Table 8: Clustering scores of RoBERTa-base trained with ER in the CIL scenario. The last row reports the trained head's A_T, the average accuracy over all tasks after learning the final task T.

Algorithm	Metric	TACRED	Banking77	CLINC150	FewRel	Topic3
K-means	ACC	52.04 ± 1.5	65.23 ± 2.0	78.37 ± 1.8	62.29 ± 2.2	66.50 ± 1.7
	ARI	54.90 ± 1.3	71.68 ± 1.7	83.91 ± 2.0	66.18 ± 1.5	74.55 ± 1.8
	NMI	71.71 ± 1.4	80.55 ± 1.9	87.32 ± 2.1	77.91 ± 2.0	84.35 ± 1.6
GMM	ACC	52.06 ± 1.6	65.77 ± 1.8	78.90 ± 2.0	61.44 ± 2.1	61.70 ± 1.7
	ARI	57.62 ± 1.5	74.76 ± 1.9	81.73 ± 2.1	64.29 ± 1.8	73.53 ± 1.6
	NMI	71.15 ± 1.3	83.75 ± 2.0	88.40 ± 2.2	76.90 ± 1.9	82.16 ± 1.5
Spectral	ACC	59.82 ± 1.4	67.92 ± 2.0	80.02 ± 1.7	60.18 ± 1.9	61.80 ± 1.8
	ARI	55.40 ± 1.5	72.18 ± 1.8	83.18 ± 2.1	60.40 ± 2.0	76.23 ± 1.6
	NMI	70.17 ± 1.6	82.95 ± 1.9	89.47 ± 2.2	75.73 ± 2.0	84.68 ± 1.7
Agglomerative	ACC	63.82 ± 1.7	70.81 ± 1.8	79.69 ± 2.1	63.94 ± 1.9	67.28 ± 1.6
	ARI	61.30 ± 1.4	75.57 ± 2.0	84.34 ± 2.2	67.95 ± 1.8	77.96 ± 1.7
	NMI	72.20 ± 1.5	85.12 ± 2.1	90.57 ± 2.3	78.55 ± 2.0	84.56 ± 1.8
Deep Clustering	ACC	59.58 ± 1.6	67.36 ± 1.9	80.45 ± 2.1	63.18 ± 1.7	61.44 ± 1.8
	ARI	61.62 ± 1.5	74.27 ± 1.8	84.02 ± 2.0	66.50 ± 1.6	73.54 ± 1.9
	NMI	70.51 ± 1.4	83.44 ± 2.0	88.79 ± 2.3	78.11 ± 2.1	83.66 ± 1.7
Trained head	A_T	49.19 ± 1.3	52.79 ± 1.8	66.69 ± 2.0	42.54 ± 1.9	73.23 ± 1.6

ARI [Hubert and Arabie, 1985] measures the agreement between a clustering and the ground-truth labels over pairs of data points, corrected for chance. It equals 1 for perfect agreement and 0 in expectation for a random clustering; negative values indicate agreement below chance [Vinh et al., 2010, Zhang et al., 2019]. With the pair counts defined below, ARI can be written as

$$ARI = \frac{2 \cdot (TP \cdot TN - FN \cdot FP)}{(TP + FN) \cdot (FN + TN) + (TP + FP) \cdot (FP + TN)} \tag{5}$$

TP (true positive): the number of sample pairs placed in the same cluster that also share the same ground-truth label.

TN (true negative): the number of sample pairs placed in different clusters that also have different labels.

FP (false positive): the number of sample pairs placed in the same cluster but with different labels.

FN (false negative): the number of sample pairs placed in different clusters but with the same label.

NMI (Normalized Mutual Information)

NMI measures the mutual information between the cluster assignments and the ground-truth labels, normalized by their entropies. It ranges from 0 to 1, with values closer to 1 indicating stronger dependence between clusters and labels.

$$I(U, V) = \sum_{u \in U} \sum_{v \in V} P(u, v) \cdot \log \left(\frac{P(u, v)}{P(u) \cdot P(v)} \right) \tag{6}$$

$$H(U) = - \sum_{u \in U} P(u) \cdot \log P(u) \tag{7}$$

$$NMI(U, V) = \frac{I(U, V)}{\sqrt{H(U) H(V)}} \tag{8}$$

Here U denotes the cluster assignment and V the ground-truth labels, $I(U, V)$ is their mutual information, and $H(\cdot)$ is entropy ($H(V)$ is defined analogously to $H(U)$). NMI normalizes $I(U, V)$ by the geometric mean of $H(U)$ and $H(V)$.

Table 9: Clustering scores of Pythia-410M trained with LAMOL_g in the CIL scenario. The last row reports the trained head's A_T, the average accuracy over all tasks after learning the final task T.

Algorithm	Metric	TACRED	Banking77	CLINC150	FewRel	Topic3
K-means	ACC	27.36 ± 3.2	67.13 ± 2.8	71.39 ± 4.5	57.23 ± 3.1	63.23 ± 2.4
	ARI	24.66 ± 2.6	71.23 ± 3.1	76.24 ± 4.2	62.12 ± 2.9	68.22 ± 2.7
	NMI	45.67 ± 2.3	81.93 ± 3.0	85.25 ± 3.8	74.33 ± 4.5	80.91 ± 3.1
GMM	ACC	27.77 ± 2.8	65.38 ± 3.4	71.89 ± 3.7	58.36 ± 4.2	59.27 ± 2.5
	ARI	23.97 ± 3.1	70.46 ± 2.6	77.82 ± 4.1	61.99 ± 3.7	72.23 ± 3.0
	NMI	46.03 ± 3.4	82.23 ± 2.9	84.91 ± 3.3	73.10 ± 4.4	81.85 ± 3.5
Spectral	ACC	28.74 ± 4.1	66.40 ± 2.5	71.44 ± 3.6	55.13 ± 3.9	48.80 ± 2.8
	ARI	21.46 ± 3.7	69.45 ± 2.7	76.61 ± 4.4	56.08 ± 4.3	73.90 ± 3.1
	NMI	46.38 ± 3.2	82.42 ± 2.8	86.06 ± 4.1	72.96 ± 3.5	84.48 ± 2.7
Agglomerative	ACC	28.30 ± 3.1	68.12 ± 2.6	72.16 ± 3.8	61.79 ± 3.7	76.09 ± 3.9
	ARI	24.00 ± 2.7	71.44 ± 3.5	78.11 ± 3.3	65.40 ± 3.1	80.59 ± 4.2
	NMI	46.99 ± 2.9	83.86 ± 2.4	87.65 ± 4.0	76.31 ± 4.1	83.22 ± 3.5
Deep Clustering	ACC	28.12 ± 2.8	66.32 ± 3.6	71.23 ± 4.2	56.12 ± 3.5	65.42 ± 4.3
	ARI	24.34 ± 3.3	69.23 ± 3.4	77.24 ± 3.7	63.78 ± 4.1	71.10 ± 3.6
	NMI	46.23 ± 2.6	83.23 ± 3.5	86.11 ± 3.9	74.60 ± 4.2	82.12 ± 3.3
Trained head	A_T	28.95 ± 3.2	51.95 ± 4.3	34.38 ± 3.1	23.09 ± 4.2	74.65 ± 3.4

B.2 Experiment results by clustering algorithm

We evaluated all models and IL methods with five clustering algorithms and three metrics. Most clustering algorithms show broadly similar patterns. For comparison with the commonly reported Average Accuracy of the trained head, we present the results of experiments with ER method on RoBERTa-base model as a representative of discriminative backbone in Table 8. Similarly, we present the results of experiments with LAMOL_g method on Pythia-410M model as a representative of generative backbone in Table 9. A_T is the trained head's Average Accuracy over all tasks after the final task T (Appendix A.3.1).

In the CIL scenario, the clustering scores are often higher than the trained head's A_T. For RoBERTa-base with ER (Table 8), clustering ACC exceeds A_T for all five algorithms on the four datasets with seven or more incremental tasks (TACRED, Banking77, CLINC150, and FewRel), whereas on Topic3Datasets, which has only five tasks, A_T is higher than the ACC of every algorithm. For Pythia-410M with LAMOL_g (Table 9), clustering ACC exceeds A_T on Banking77, CLINC150, and FewRel but is close to or slightly below A_T on TACRED. This pattern suggests that as the number of incremental tasks increases, head-level average accuracy can become increasingly affected by forgetting in the head. The clustering results should not be interpreted as higher task accuracy, but as evidence that representation separability can remain higher than what the trained head exposes.

Our goal in this study is not to determine which clustering algorithm is best or which metric is the best. We therefore use Spectral Clustering, whose scores lie in the middle of the range, for the visualizations and analysis in the main paper; all results are reported in Appendix E.

B.3 Sensitivity of the clustering diagnostic

Using the features stored for the 27 additional units, we recompute the clustering diagnostic under 25 settings. Starting from the default (Spectral, no normalization, $k = C$, seed 0, Hungarian matching, natural class distribution, $n = 1{,}500$), each setting changes one factor: normalization (L2, standardization, whitening), number of clusters ($0.5C$, $2C$), seed (1, 2), label matching (majority vote), class balance (balanced subsampling), sample size (750, 3,000, full test set), and algorithm (K-means, GMM, Agglomerative); we also cross the three alternative algorithms with the four normalizations. Because all settings use identical stored features, differences reflect diagnostic choices rather than training variation. Table 10 shows that the BWT–FWT correlation is negative in every setting for ACC, ARI, and NMI.

Table 10: Sensitivity of the clustering-based backbone diagnostic: BWT–FWT Pearson correlation over the 27 additional units under 25 settings, each recomputed offline from the same saved features. The default uses Spectral Clustering without normalization, $k = C$ clusters, seed 0, Hungarian matching, the natural class distribution, and $n = 1{,}500$ samples; each row in the upper block changes one factor, and the lower block crosses three algorithms with four normalizations.

Factor	**Setting**	**ACC**	**ARI**	**NMI**
Default	Spectral, no norm., k=C, seed 0, Hungarian	−0.67	−0.70	−0.58
Normalization	L2	−0.67	−0.66	−0.52
	Standardization	−0.64	−0.65	−0.50
	Whitening	−0.63	−0.57	−0.51
Number of clusters	$k = 0.5C$	−0.82	−0.63	−0.48
	$k = 2C$	−0.81	−0.82	−0.60
Seed	1	−0.67	−0.71	−0.57
	2	−0.70	−0.75	−0.61
Label matching	Majority vote	−0.75	−0.70	−0.58
Class balance	Balanced subsampling	−0.62	−0.68	−0.50
Sample size	$n = 750$	−0.65	−0.68	−0.42
	$n = 3{,}000$	−0.63	−0.64	−0.60
	Full test set	−0.66	−0.62	−0.61
Algorithm × normalization				
K-means	None	−0.60	−0.57	−0.51
	L2	−0.57	−0.61	−0.55
	Standardization	−0.62	−0.60	−0.51
	Whitening	−0.61	−0.39	−0.47
GMM	None	−0.58	−0.55	−0.47
	L2	−0.56	−0.60	−0.52
	Standardization	−0.49	−0.58	−0.59
	Whitening	−0.46	−0.26	−0.38
Agglomerative	None	−0.62	−0.60	−0.58
	L2	−0.66	−0.38	−0.56
	Standardization	−0.60	−0.58	−0.57
	Whitening	−0.62	−0.38	−0.32

C Incremental learning methods

We measured BWT and FWT for representative Incremental Learning methods. Aside from a brief explanation, we adopted detailed experimental settings widely used in prior studies.

Base - Sequentially fine-tunes the model on each task without any mechanism against forgetting. When evaluated through the trained head, it tends to predict mostly the classes of the most recently trained task.

ER [Chaudhry et al., 2019] - A classical replay technique that incorporates old samples to improve retention. When learning a new task, a portion of old samples is included in the training process. In this study, we include one old sample per class during training.

DER++ [Buzzega et al., 2020] - DER++ extends ER by also storing the model's logits for buffered samples and matching them with an MSE loss, in addition to replaying their labels. Although originally developed for the computer vision domain, we evaluated this method with both discriminative and generative backbones.

CLSER [Arani et al., 2022] - CLS-ER extends experience replay with two semantic memories, a plastic and a stable model maintained as exponential moving averages of the working model, and regularizes the working model toward them. Like DER++, this method was initially used in the computer vision domain, but we applied it to both discriminative and generative backbones.

L2KD [Chuang et al., 2020] - L2KD builds on LAMOL and adds knowledge distillation: for each new task, a teacher model is first trained on that task and then distilled into the lifelong learner.

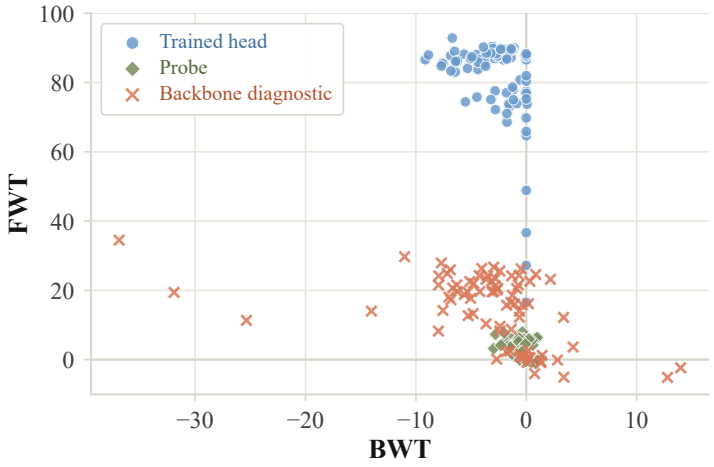


Figure 4: TIL sanity check. The x-axis represents BWT and the y-axis represents operational FWT; colors denote the three evaluation interfaces. Even when task identity restricts the prediction space, the interfaces remain separated, supporting that the backbone–head gap is not only CIL classifier bias.

LAMOL_g & LAMOL_t [Sun et al., 2020] - LAMOL is a method designed for generative models. When training on a new task, the model generates pseudo-samples of previous tasks and learns them together with the new task. The two variants differ in the generation token: LAMOL_g uses a single shared generation token for all tasks, whereas LAMOL_t uses a task-specific token.

LAMOL_KD [Zheng et al., 2024] - Similar to L2KD, except that the teacher model is trained on all previous tasks and transfers this knowledge through knowledge distillation.

PCLL [Zhao et al., 2022] - PCLL builds on LAMOL-style generative replay, using a prompt-conditioned variational autoencoder (VAE) to generate pseudo samples of previous tasks, combined with knowledge distillation.

SEQ* [Zheng et al., 2024] - SEQ* reduces backbone forgetting and classifier bias by freezing the backbone after warming up on the first task and training only the classifier for subsequent tasks using frozen backbone outputs. Since the backbone remains fixed after the first task, its representation-level diagnostic scores do not change. This makes SEQ* useful as a frozen-backbone baseline, but its BWT-backbone and FWT-backbone are always 0 by construction.

KLDA [Momeni et al., 2025] - KLDA projects input data into a high-dimensional feature space using a kernel function and then performs Linear Discriminant Analysis (LDA) to maximize class separability. It operates on fixed embeddings extracted from a foundation model, so the backbone is not updated during continual learning. We report its ensemble variant, KLDA-E.

D Results in the TIL scenario

The interface discrepancy in the CIL results could in principle arise only from class-incremental classifier bias. The TIL setting tests this explanation, because task identity restricts the prediction space. Figure 4 shows that the interface discrepancy remains: head-level and probe-level evaluations become less dominated by global class competition, but the backbone diagnostic still shows the most consistent stability–plasticity pattern among the evaluated interfaces. The backbone–head gap is therefore broader than class-incremental classifier bias.

We compare BWT and FWT under the three interfaces using four Incremental Learning methods across four discriminative backbones, as in the CIL scenario. As shown in Table 11 (and in Figure 5 for RoBERTa-base), the backbone diagnostic yields the strongest negative BWT–FWT correlation for all four backbones. Even in the TIL scenario, where a separate classifier is used for each task, the stability–plasticity pattern appears more clearly in the backbone diagnostic than in the trained head.

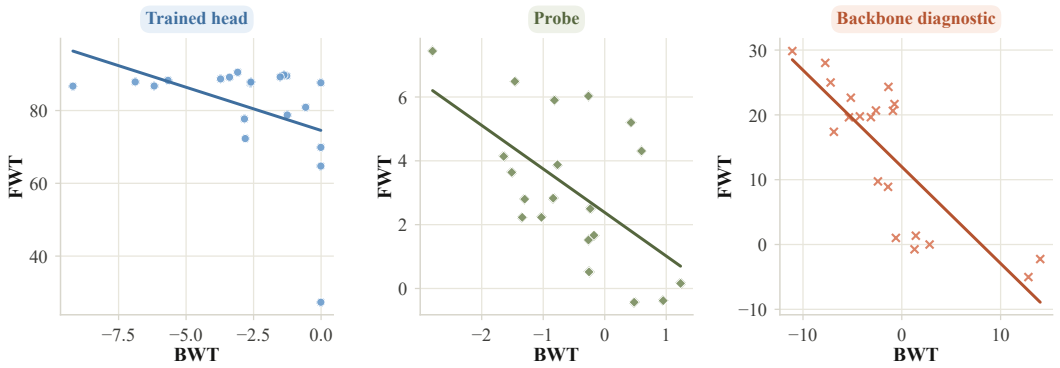
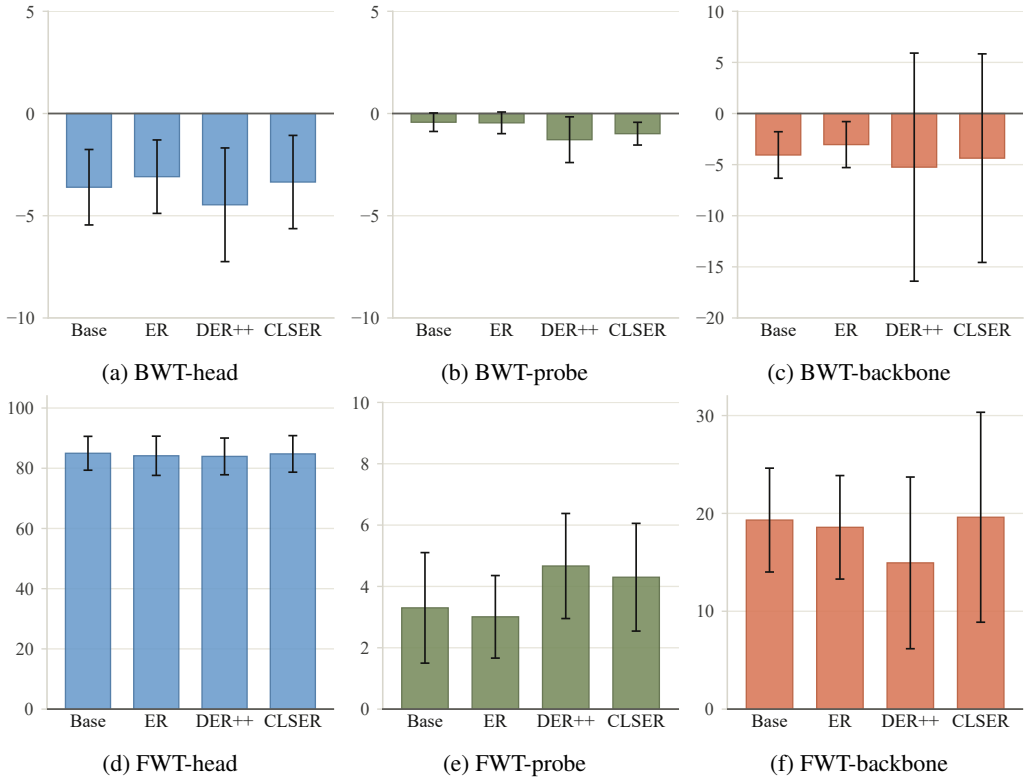


Figure 5: BWT versus operational FWT under the three interfaces for RoBERTa-base in the TIL scenario (Base, ER, DER++, and CLSER), with least-squares fits. Correlation coefficients are reported in Table 11.



(a) BWT-head (b) BWT-probe (c) BWT-backbone

(d) FWT-head (e) FWT-probe (f) FWT-backbone

Figure 6: Per-method BWT and operational FWT of the discriminative backbones in the TIL scenario under the three interfaces (mean and standard deviation).

Observing the x-axis in Figure 4, BWT-head values are close to 0 in the TIL scenario, as separate classifiers are used for each task. However, prior research [Zhou and Srikumar, 2021] has suggested that classifiers can account for much of the performance, even when the model's representation is relatively weak. In comparison, BWT-backbone spans a much broader range of values, reflecting differences in representation change across models and IL methods that are less visible through head-level accuracy. BWT-probe, as in the CIL scenario, is close to 0 for all models and methods because the probe is retrained for each evaluation.

On the y-axis of Figure 4, the operational FWT results for the three interfaces show distinct differences even in the TIL scenario. FWT-head values exceed 60 in most cases, as they are measured on heads that have no outputs for the next classes, similar to the CIL scenario. Lastly, FWT-backbone

Table 11: Pearson correlation coefficient between BWT and operational FWT for each interface and clustering metric in the TIL scenario. The results aggregate the benchmark streams in Table 6 for four methods: Base, ER, DER++, and CLSER.

		Discriminative backbones			
Interface	**Metric**	BERT-base	BERT-large	RoBERTa-base	RoBERTa-large
Trained head	ACC	−0.46	−0.61	−0.51	−0.63
Probe	ACC	−0.55	−0.50	−0.47	−0.62
Backbone diagnostic					
K-means	ACC	−0.84	−0.80	−0.87	−0.81
	ARI	−0.88	−0.81	−0.80	−0.87
	NMI	−0.88	−0.83	−0.84	−0.82
GMM	ACC	−0.80	−0.88	−0.87	−0.78
	ARI	−0.89	−0.88	−0.80	−0.85
	NMI	−0.88	−0.85	−0.82	−0.84
Spectral	ACC	−0.81	−0.87	−0.83	−0.80
	ARI	−0.87	−0.85	−0.89	−0.87
	NMI	−0.85	−0.82	−0.84	−0.83
Agglomerative	ACC	−0.87	−0.85	−0.80	−0.89
	ARI	−0.91	−0.88	−0.82	−0.88
	NMI	−0.89	−0.84	−0.86	−0.80
Deep Clustering	ACC	−0.81	−0.79	−0.88	−0.83
	ARI	−0.88	−0.82	−0.84	−0.82
	NMI	−0.88	−0.85	−0.81	−0.89

suggests performance improvements in the backbone as new tasks are learned, which aligns with the explanation provided for the CIL scenario in Section 3.3.

In Figures 6a and 6c, ER shows the least forgetting and DER++ the most under both the trained head and the backbone diagnostic; the two interfaces differ only in the order of Base and CLSER, whose values lie within 0.35 of each other in both cases. BWT-probe (Figure 6b) ranks the methods differently, as expected for a probe trained separately from the backbone. In the TIL scenario, task-specific classifiers reduce global class-incremental classifier bias, but the trained head and the probe still depend on their readout mechanisms. BWT-backbone ranks the methods almost identically to BWT-head in the TIL scenario, suggesting that separability-based forgetting can be approximated without using the trained head.

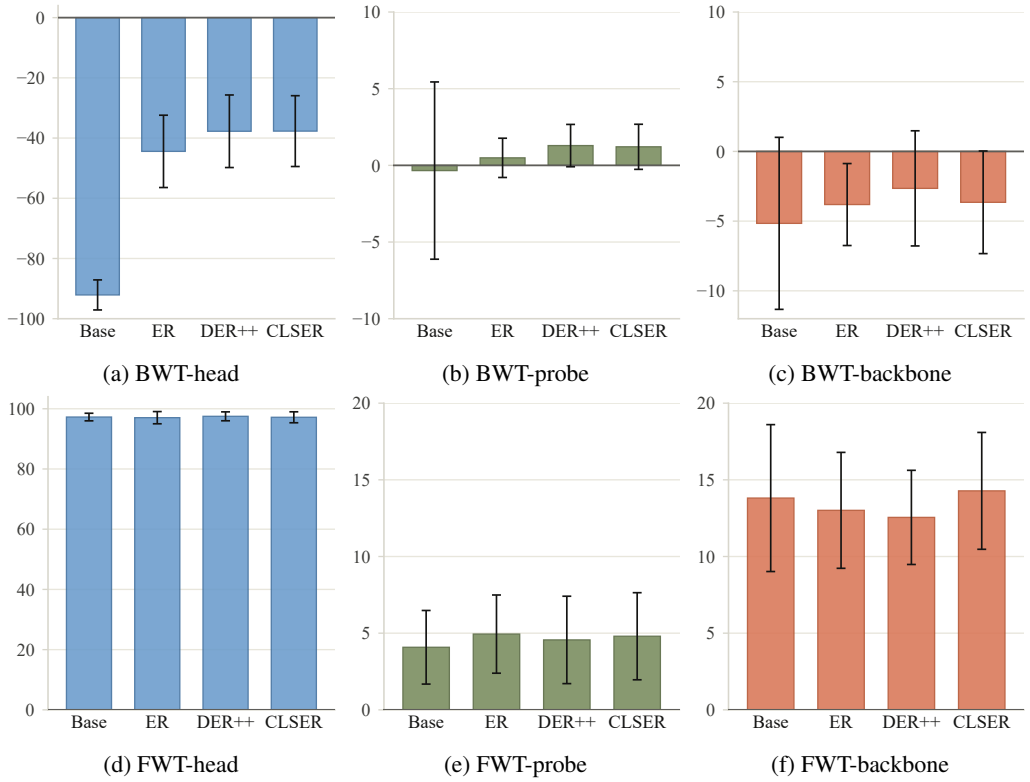
Figures 6d, 6e, and 6f compare the operational FWT measurement results in the TIL scenario. In Figure 6d, FWT-head scores all averaged above 80, showing little difference across methods. In Figure 6e, FWT-probe recorded extremely low values, averaging below 5 for all methods, with no discernible distinction. In Figure 6f, similar to Figure 7f, the results resemble those of the CIL scenario, as the backbone undergoes the same processes except for the difference in the classifiers used for CIL and TIL. This supports our interpretation that FWT-backbone measures backbone plasticity more independently of the trained head. The corresponding correlation coefficients for all four backbones are reported in Table 11.

We report TIL results for discriminative backbones. Most generative IL methods are designed for the CIL setting, and in TIL the Base method already performs well because each task has its own prediction space.

E Full results

E.1 Per-method results and correlations across clustering choices

Figure 7 shows the per-method CIL bar plots for the discriminative backbones, and Figure 3 in the main text shows those for the generative backbones. They show the same qualitative pattern described in Figure 2 and Section 5.1: the trained head suggests severe forgetting, the probe suggests high recoverability, and the backbone diagnostic lies between these two views.



(a) BWT-head (b) BWT-probe (c) BWT-backbone

(d) FWT-head (e) FWT-probe (f) FWT-backbone

Figure 7: Average and standard deviation of head-level, probe-level, and backbone-level BWT/FWT for each IL method in four discriminative backbones, aggregated over the benchmark streams in Table 6.

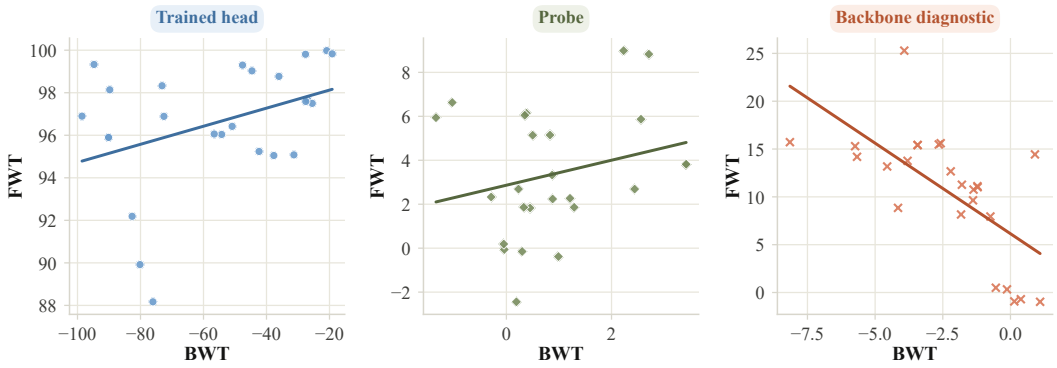


Figure 8: BWT versus operational FWT under the three interfaces for RoBERTa-base in the CIL scenario, with least-squares fits. Correlation coefficients are reported in Tables 12 and 13.

We conducted the same experiment for all clustering algorithms and metrics other than those analyzed in the main paper. For the discriminative backbones in Tables 12 and 13, the BWT–FWT correlations measured through the trained head and the probe are positive (0.11 to 0.65), whereas the backbone diagnostic is negative in every configuration. This suggests that the stability–plasticity pattern is interface-dependent rather than a direct consequence of the metric formula.

The backbone diagnostic is negative regardless of whether ACC, ARI, or NMI is used. For the generative backbones, the trained-head and probe correlations are also negative (-0.36 to -0.62) but weaker than the backbone diagnostic (-0.70 to -0.93). Because the same shared-term structure applies to every interface, we use this pattern as comparative diagnostic evidence rather than as standalone proof of a universal trade-off.

Table 12: Pearson correlation coefficient between BWT and operational FWT for each interface and clustering metric in the CIL scenario.

		Discriminative backbones				Generative backbones					
Interface	**Metric**	BERT base	BERT large	RoBERTa base	RoBERTa large	Pythia 70M	Pythia 160M	Pythia 410M	Qwen2 0.5B	Qwen2.5 0.5B	Qwen3 0.6B
Trained head	ACC	0.43	0.14	0.38	0.24	−0.50	−0.36	−0.37	−0.52	−0.36	−0.38
Probe	ACC	0.65	0.11	0.25	0.11	−0.42	−0.48	−0.62	−0.43	−0.48	−0.62
Backbone diagnostic											
K-means	ACC	−0.63	−0.56	−0.48	−0.52	−0.85	−0.87	−0.85	−0.85	−0.87	−0.86
	ARI	−0.66	−0.51	−0.53	−0.54	−0.80	−0.81	−0.82	−0.80	−0.81	−0.82
	NMI	−0.68	−0.61	−0.58	−0.68	−0.82	−0.91	−0.89	−0.82	−0.91	−0.90
GMM	ACC	−0.52	−0.57	−0.50	−0.43	−0.77	−0.75	−0.77	−0.77	−0.75	−0.78
	ARI	−0.68	−0.69	−0.62	−0.51	−0.81	−0.80	−0.84	−0.81	−0.80	−0.85
	NMI	−0.69	−0.70	−0.61	−0.61	−0.79	−0.83	−0.84	−0.79	−0.84	−0.85
Spectral	ACC	−0.69	−0.61	−0.62	−0.60	−0.81	−0.83	−0.87	−0.81	−0.83	−0.86
	ARI	−0.68	−0.62	−0.68	−0.65	−0.82	−0.85	−0.81	−0.83	−0.85	−0.82
	NMI	−0.72	−0.64	−0.66	−0.69	−0.82	−0.89	−0.87	−0.83	−0.89	−0.87
Agglomerative	ACC	−0.47	−0.51	−0.61	−0.46	−0.77	−0.75	−0.70	−0.77	−0.75	−0.70
	ARI	−0.74	−0.69	−0.60	−0.74	−0.87	−0.81	−0.89	−0.86	−0.82	−0.89
	NMI	−0.91	−0.71	−0.73	−0.75	−0.88	−0.93	−0.87	−0.88	−0.92	−0.88
Deep Clustering	ACC	−0.50	−0.49	−0.62	−0.59	−0.79	−0.82	−0.78	−0.80	−0.82	−0.88
	ARI	−0.47	−0.52	−0.59	−0.48	−0.84	−0.81	−0.80	−0.84	−0.81	−0.81
	NMI	−0.63	−0.64	−0.67	−0.62	−0.87	−0.83	−0.88	−0.87	−0.83	−0.89

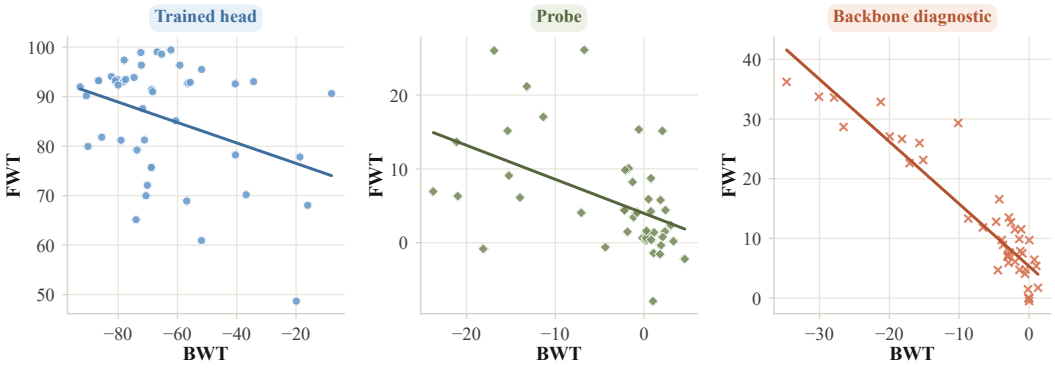


Figure 9: BWT versus operational FWT under the three interfaces for Pythia-160M in the CIL scenario, with least-squares fits. Correlation coefficients are reported in Tables 12 and 13.

E.2 Additional units and clustering-free diagnostics

The additional units comprise Llama-3.2-1B with full fine-tuning on TACRED, Banking77, CLINC150, and FewRel × Base, DER++, and CLSER (12 units); Llama-3.2-1B with LoRA and prompt tuning on TACRED, Banking77, and CLINC150 (6 units); and Qwen3-0.6B with full fine-tuning on TACRED, Banking77, and CLINC150 × Base, DER++, and CLSER (9 units). At every task boundary, we store the backbone features of the test sets of all tasks, including future tasks, so that every diagnostic is computed offline from the same features. NCM and k-nearest-neighbor accuracy ($k \in \{1, 10\}$) are measured on a fixed stratified 50/50 split of the test features, with class means or neighbors taken from one half and accuracy measured on the other; the silhouette score uses cosine distance and ground-truth labels ($\times 100$), and the Fisher ratio is $100\, S_b/(S_b + S_w)$ for between-class scatter S_b and within-class scatter S_w. These diagnostics use L2-normalized features. The clustering diagnostic uses Spectral Clustering without normalization, $k = C$ clusters, seed 0, Hungarian matching, the natural class distribution, and $n = 1{,}500$ samples. Table 14 reports all units.

Table 13: Spearman correlation coefficient between BWT and operational FWT for each interface and clustering metric in the CIL scenario.

		Discriminative backbones				Generative backbones					
Interface	**Metric**	BERT base	BERT large	RoBERTa base	RoBERTa large	Pythia 70M	Pythia 160M	Pythia 410M	Qwen2 0.5B	Qwen2.5 0.5B	Qwen3 0.6B
Trained head	ACC	0.42	0.13	0.38	0.24	−0.51	−0.37	−0.37	−0.53	−0.38	−0.37
Probe	ACC	0.64	0.11	0.24	0.10	−0.42	−0.47	−0.63	−0.43	−0.49	−0.63
Backbone diagnostic											
K-means	ACC	−0.62	−0.56	−0.49	−0.52	−0.84	−0.87	−0.85	−0.84	−0.86	−0.85
	ARI	−0.66	−0.52	−0.53	−0.54	−0.81	−0.80	−0.83	−0.81	−0.81	−0.83
	NMI	−0.65	−0.61	−0.58	−0.67	−0.82	−0.92	−0.89	−0.82	−0.91	−0.89
GMM	ACC	−0.51	−0.58	−0.49	−0.43	−0.76	−0.74	−0.78	−0.77	−0.75	−0.78
	ARI	−0.67	−0.69	−0.62	−0.52	−0.82	−0.81	−0.85	−0.82	−0.81	−0.86
	NMI	−0.55	−0.59	−0.61	−0.61	−0.79	−0.84	−0.84	−0.79	−0.83	−0.84
Spectral	ACC	−0.69	−0.61	−0.61	−0.60	−0.82	−0.83	−0.86	−0.82	−0.83	−0.86
	ARI	−0.68	−0.63	−0.68	−0.65	−0.82	−0.85	−0.81	−0.82	−0.85	−0.81
	NMI	−0.62	−0.64	−0.65	−0.61	−0.82	−0.88	−0.88	−0.82	−0.88	−0.87
Agglomerative	ACC	−0.47	−0.51	−0.61	−0.46	−0.78	−0.74	−0.70	−0.79	−0.74	−0.71
	ARI	−0.74	−0.69	−0.60	−0.73	−0.86	−0.81	−0.89	−0.87	−0.81	−0.89
	NMI	−0.90	−0.71	−0.73	−0.62	−0.88	−0.92	−0.88	−0.88	−0.92	−0.88
Deep Clustering	ACC	−0.50	−0.49	−0.61	−0.58	−0.80	−0.83	−0.89	−0.81	−0.83	−0.89
	ARI	−0.48	−0.53	−0.58	−0.49	−0.85	−0.82	−0.79	−0.85	−0.82	−0.80
	NMI	−0.52	−0.54	−0.60	−0.63	−0.87	−0.72	−0.79	−0.88	−0.72	−0.79

E.3 Parameter-efficient methods

Table 15 reports the six parameter-efficient units summarized in Section 5.3. For prompt tuning, head accuracy remains low throughout (A_T 4.6–10.2), so its comparatively small BWT-head reflects low accuracy rather than retention.

E.4 Full JLT results

Tables 16 and 17 report the full per-model and per-dataset JLT results corresponding to the aggregate gap summarized in Table 3.

E.5 Model size & capacity

Figure 10 (Section 5.4) reports BWT-backbone, FWT-backbone, and their sum for the Base method in three groups of representative LLMs (GPT-style Pythia, Llama, and Qwen). The sum $BWT + FWT = a_{T,i} - a_{i-1,i}$ cancels the shared post-training term $a_{i,i}$, so we treat it as a coupling-aware net-transfer view rather than as a primary metric. The sum increases with model size within each model family. In particular, BWT-backbone improves from the smallest to the largest model in every family, indicating a positive relationship between model size and retention. On the other hand, FWT-backbone changes little across models and sizes. The relatively stable operational FWT despite larger model sizes should be considered when designing future IL methods.

F LM-head adaptation details

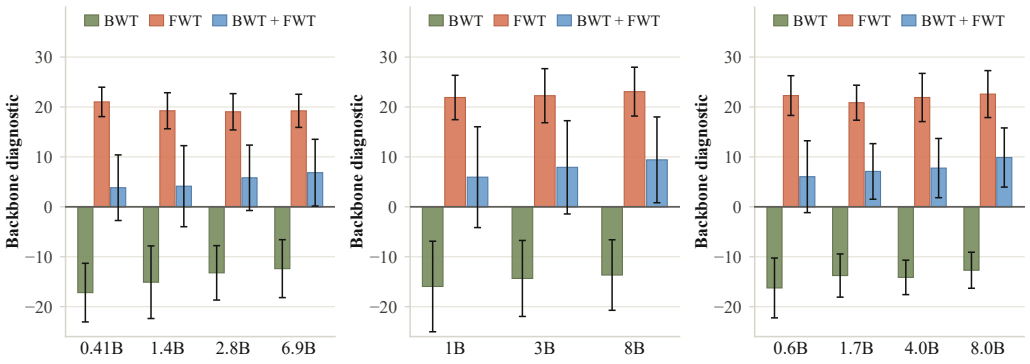
Setup. For the JLT diagnostic, the backbone is fixed after sequential IL training and only the LM head is updated. We use two head-adaptation modes. In post-hoc JLT, the head is re-aligned once per checkpoint with a small replay buffer and no task-specific adaptation (Section F.2). In task-adaptive JLT, the backbone is frozen and the LM head is briefly adapted on the training split of the current task before that task is evaluated (see Section F.1). The latter is an oracle-style recovery diagnostic and is illustrated in Figure 11.

Table 14: Additional units: BWT (top) and operational FWT (bottom) under each interface. Llama: Llama-3.2-1B; Qwen: Qwen3-0.6B. Spectral is the clustering-based diagnostic in its default setting; NCM, kNN, silhouette (Silh.), and Fisher ratio are clustering-free. A_T is the trained head's final average accuracy. The last row reports the BWT–FWT Pearson correlation over the 27 units.

					Backbone diagnostic					
Backbone	**Dataset**	**Method**	A_T	**Head**	**Spectral**	**NCM**	**kNN-1**	**kNN-10**	**Silh.**	**Fisher**
BWT										
Llama	Banking77	LoRA	14.90	−86.06	−1.91	−3.94	−1.06	−2.95	−7.05	−3.39
Llama	Banking77	Prompt tuning	5.23	−18.64	4.11	1.06	2.58	0.61	−2.83	−4.37
Llama	CLINC150	LoRA	12.02	−88.95	−8.90	−3.81	−3.33	−4.19	−8.14	−5.73
Llama	CLINC150	Prompt tuning	10.20	−49.12	−5.04	−1.33	−0.76	−1.19	−2.76	−2.38
Llama	TACRED	LoRA	16.00	−76.54	−10.74	−6.32	−1.07	−6.41	−4.08	−6.92
Llama	TACRED	Prompt tuning	4.59	−50.31	−4.38	−6.66	−4.67	−4.01	−1.62	−1.65
Llama	Banking77	CLSER	15.23	−91.29	−1.02	−9.39	−0.83	−8.18	−11.60	−2.56
Llama	Banking77	DER++	14.58	−93.26	−7.22	−12.58	−4.55	−8.71	−15.58	−7.17
Llama	Banking77	Base	26.98	−82.35	−3.71	−2.27	−0.76	−1.82	−6.88	−4.64
Llama	CLINC150	CLSER	6.09	−77.93	−7.09	−11.00	−3.14	−5.71	−7.65	0.53
Llama	CLINC150	DER++	6.09	−80.88	−7.11	−12.71	−1.57	−8.29	−10.61	−2.09
Llama	CLINC150	Base	47.56	−55.36	−5.63	−1.57	−1.76	−2.57	−4.63	−2.43
Llama	FewRel	CLSER	46.31	−58.14	−0.32	−1.92	1.37	−0.37	−5.11	−4.88
Llama	FewRel	DER++	48.40	−54.96	1.97	0.51	1.80	0.96	−1.19	−0.46
Llama	FewRel	Base	37.80	−68.38	0.41	−2.88	−0.12	−1.49	−8.19	−6.53
Llama	TACRED	CLSER	25.99	−80.69	3.05	3.29	2.63	1.41	0.58	−1.76
Llama	TACRED	DER++	35.44	−69.18	−3.75	−2.08	−2.78	−3.58	−5.25	−3.78
Llama	TACRED	Base	33.68	−72.07	−14.57	−5.99	−9.95	−9.57	−10.84	−11.32
Qwen	Banking77	CLSER	12.63	−88.71	−17.59	−36.36	−21.97	−24.70	−42.12	−26.20
Qwen	Banking77	DER++	13.47	−87.39	−12.53	−25.53	−10.68	−18.41	−31.87	−16.51
Qwen	Banking77	Base	24.54	−84.89	−0.95	−1.89	−0.30	−2.80	−12.86	−9.40
Qwen	CLINC150	CLSER	6.44	−93.95	−22.51	−36.43	−17.57	−26.19	−44.92	−21.96
Qwen	CLINC150	DER++	6.42	−92.74	−24.38	−35.24	−17.71	−24.90	−42.16	−21.32
Qwen	CLINC150	Base	45.51	−57.48	−4.40	−2.86	−2.33	−4.14	−17.62	−14.12
Qwen	TACRED	CLSER	16.35	−90.20	−6.36	−21.86	−9.64	−16.01	−10.88	−6.44
Qwen	TACRED	DER++	17.70	−88.97	−1.87	−8.88	−7.82	−4.47	−5.18	−9.25
Qwen	TACRED	Base	21.07	−85.17	−9.35	−15.91	−5.73	−13.01	−9.34	−10.97
Operational FWT										
Llama	Banking77	LoRA	–	87.54	2.70	6.29	1.52	2.65	6.66	9.05
Llama	Banking77	Prompt tuning	–	17.65	−0.73	2.35	3.26	2.65	4.19	3.94
Llama	CLINC150	LoRA	–	90.90	6.53	2.38	1.62	1.76	7.30	5.92
Llama	CLINC150	Prompt tuning	–	48.31	7.73	1.71	1.24	1.14	4.18	3.46
Llama	TACRED	LoRA	–	80.75	2.37	4.10	0.96	4.86	0.53	1.82
Llama	TACRED	Prompt tuning	–	47.52	2.76	4.29	0.92	3.67	0.74	−0.40
Llama	Banking77	CLSER	–	93.45	16.89	19.24	8.48	14.09	30.18	20.05
Llama	Banking77	DER++	–	94.70	14.13	20.45	6.74	14.02	31.21	22.27
Llama	Banking77	Base	–	95.80	6.06	6.67	3.41	4.32	14.16	9.76
Llama	CLINC150	CLSER	–	83.81	14.31	31.05	18.14	22.67	43.08	29.73
Llama	CLINC150	DER++	–	86.76	13.60	28.81	15.90	21.38	43.99	26.20
Llama	CLINC150	Base	–	89.86	6.33	2.81	2.38	4.19	11.78	8.53
Llama	FewRel	CLSER	–	91.59	9.38	8.73	4.63	6.27	12.57	10.43
Llama	FewRel	DER++	–	91.37	2.61	5.43	2.59	4.61	6.50	5.02
Llama	FewRel	Base	–	90.05	9.66	9.37	4.73	5.80	16.27	11.74
Llama	TACRED	CLSER	–	95.84	−1.35	3.46	5.24	4.21	4.22	6.11
Llama	TACRED	DER++	–	95.43	−0.38	2.35	3.22	2.64	1.14	5.03
Llama	TACRED	Base	–	96.22	4.36	3.61	−1.40	3.42	4.37	6.49
Qwen	Banking77	CLSER	–	88.90	13.30	37.05	15.68	21.67	45.13	22.80
Qwen	Banking77	DER++	–	88.56	11.01	29.92	13.26	20.30	42.48	23.19
Qwen	Banking77	Base	–	96.32	8.65	7.35	1.82	8.18	20.10	14.07
Qwen	CLINC150	CLSER	–	93.95	25.56	47.76	28.67	34.81	60.27	32.60
Qwen	CLINC150	DER++	–	92.71	27.16	44.38	25.62	33.24	55.08	27.68
Qwen	CLINC150	Base	–	96.05	10.40	5.29	3.48	7.05	22.26	14.86
Qwen	TACRED	CLSER	–	95.10	2.85	12.13	5.30	5.55	8.43	16.50
Qwen	TACRED	DER++	–	95.41	−4.87	−2.30	−3.09	−3.08	1.19	2.04
Qwen	TACRED	Base	–	95.60	11.57	11.21	10.28	14.49	14.81	16.49
Pearson r(BWT, FWT), 27 units				−0.746	−0.672	−0.872	−0.616	−0.800	−0.837	−0.497

Table 15: Parameter-efficient IL on Llama-3.2-1B (CIL). Spectral: clustering-based backbone diagnostic. NCM: nearest-class-mean accuracy on backbone features, evaluated on a fixed stratified 50/50 split of the test features (no cluster count, initialization, or cluster-label matching). The last row reports the BWT–FWT Pearson correlation over the six units.

			Head		**Spectral**		**NCM**	
Dataset	**Method**	A_T	BWT	FWT	BWT	FWT	BWT	FWT
Banking77	LoRA	14.90	−86.06	87.54	−1.91	2.70	−3.94	6.29
CLINC150	LoRA	12.02	−88.95	90.90	−8.90	6.53	−3.81	2.38
TACRED	LoRA	16.00	−76.54	80.75	−10.74	2.37	−6.32	4.10
Banking77	Prompt tuning	5.23	−18.64	17.65	4.11	−0.73	1.06	2.35
CLINC150	Prompt tuning	10.20	−49.12	48.31	−5.04	7.73	−1.33	1.71
TACRED	Prompt tuning	4.59	−50.31	47.52	−4.38	2.76	−6.66	4.29
Pearson r(BWT, FWT), six units			−0.998		−0.580		−0.563	



(a) Pythia (b) Llama (c) Qwen

Figure 10: BWT-backbone, FWT-backbone, and BWT-backbone + FWT-backbone results in Pythia, Llama, and Qwen, measured over five benchmark groups with the Base method.

Table 16: Performance recovered by head-only adaptation after incremental backbone training on three generative models. JLT rows should be interpreted as recoverable performance under the diagnostic head-adaptation protocol, not as direct replacements for standard CIL accuracy. Joint Fine-tuning denotes fine-tuning on all tasks simultaneously. A_T is defined by Equation 3 with $t = T$, BWT by Equation 1, and FWT by Equation 2. Shaded rows use the task-adaptive JLT protocol. This table covers TACRED, Banking77, and CLINC150; FewRel and Topic3 are reported in Table 17.

	TACRED			**Banking77**			**CLINC150**		
IL method	A_T	BWT	FWT	A_T	BWT	FWT	A_T	BWT	FWT
Pythia-410M									
Joint Fine-tuning	98.99	–	–	95.66	–	–	96.33	–	–
SEQ*	44.34	0.00	0.00	67.12	0.00	0.00	84.51	0.00	0.00
KLDA-E	97.20	0.00	0.00	93.03	0.00	0.00	96.62	0.00	0.00
Base	11.97	−91.26	91.28	7.56	−75.42	68.86	4.67	−89.69	87.62
L2KD	24.68	−79.27	94.39	52.32	−35.08	85.72	26.67	−73.17	94.74
LAMOL_KD	32.81	−69.55	93.68	49.48	−54.92	97.08	42.09	−60.48	97.50
PCLL	24.30	−65.29	74.37	45.91	−56.97	84.41	44.16	−57.02	91.81
Base + JLT	98.93	−0.55	11.38	75.07	−19.92	84.58	72.47	−25.50	92.29
L2KD + JLT	98.04	−0.25	4.92	95.23	−2.99	88.56	96.20	−3.38	94.52
LAMOL_KD + JLT	98.24	0.25	9.59	95.03	−3.33	88.98	96.38	−2.95	93.40
PCLL + JLT	96.70	−0.58	10.29	93.31	−3.60	87.48	96.27	−3.29	91.36
Llama-3.2-1B									
Joint Fine-tuning	98.33	–	–	96.26	–	–	97.33	–	–
SEQ*	45.82	0.00	0.00	67.48	0.00	0.00	83.72	0.00	0.00
KLDA-E	97.36	0.00	0.00	93.27	0.00	0.00	96.71	0.00	0.00
Base	12.23	−91.60	92.01	11.56	−91.10	89.28	5.89	−93.21	92.45
L2KD	31.83	−68.77	81.20	51.56	−32.46	76.52	26.44	−66.62	87.88
LAMOL_KD	32.75	−62.04	86.04	45.85	−64.91	87.01	44.80	−54.05	92.93
PCLL	31.18	−77.22	93.48	49.03	−67.69	87.40	40.69	−61.93	93.45
Base + JLT	98.63	−0.74	21.56	79.47	−17.08	87.05	74.91	−23.33	93.45
L2KD + JLT	97.06	−1.76	17.26	96.10	−2.20	87.46	96.36	−2.95	95.02
LAMOL_KD + JLT	97.15	−0.83	8.44	95.36	−2.42	84.13	87.84	−8.10	89.05
PCLL + JLT	96.30	−0.32	18.01	94.32	−2.54	87.33	96.29	−3.29	91.43
Qwen3-0.6B									
Joint Fine-tuning	98.66	–	–	93.36	–	–	96.66	–	–
SEQ*	44.09	0.00	0.00	66.84	0.00	0.00	82.91	0.00	0.00
KLDA-E	96.94	0.00	0.00	92.81	0.00	0.00	96.25	0.00	0.00
Base	11.30	−90.94	90.14	12.73	−92.39	91.44	6.60	−90.57	90.64
L2KD	34.67	−67.41	90.38	55.78	−25.99	75.11	27.31	−63.83	86.02
LAMOL_KD	42.20	−59.61	94.17	52.44	−52.31	97.42	42.71	−59.83	93.07
PCLL	49.40	−55.33	93.50	52.99	−50.68	97.78	43.27	−59.81	90.10
Base + JLT	97.49	−1.73	15.76	75.26	−22.58	86.36	77.24	−21.31	93.24
L2KD + JLT	98.21	−0.15	10.69	94.93	−3.67	86.40	95.27	−4.40	93.12
LAMOL_KD + JLT	97.88	−0.53	13.95	92.08	−6.82	84.89	90.47	−9.50	92.07
PCLL + JLT	96.23	−0.07	17.69	92.66	−5.00	87.93	95.22	−4.36	92.36

Table 17: Performance recovered by head-only adaptation on FewRel and Topic3 (continued from Table 16). Shaded rows use the task-adaptive JLT protocol.

	FewRel			Topic3		
IL method	A_T	BWT	FWT	A_T	BWT	FWT
Pythia-410M						
Joint Fine-tuning	95.35	–	–	95.50	–	–
SEQ*	61.99	0.00	0.00	70.56	0.00	0.00
KLDA-E	94.55	0.00	0.00	94.53	0.00	0.00
Base	6.17	−80.57	73.59	20.53	−91.02	93.08
L2KD	30.18	−53.23	74.57	58.17	10.27	40.25
LAMOL_KD	27.70	−75.60	95.43	50.08	−54.98	92.97
PCLL	29.79	−68.41	82.84	55.80	−42.23	92.66
Base + JLT	51.31	−36.10	77.50	72.21	−14.80	79.92
L2KD + JLT	95.18	−1.04	93.52	93.91	−0.16	92.95
LAMOL_KD + JLT	95.73	−0.54	92.82	94.00	−0.08	92.99
PCLL + JLT	94.70	−1.06	92.15	93.46	−0.26	93.51
Llama-3.2-1B						
Joint Fine-tuning	95.65	–	–	95.50	–	–
SEQ*	62.13	0.00	0.00	71.64	0.00	0.00
KLDA-E	94.68	0.00	0.00	94.59	0.00	0.00
Base	8.42	−81.39	77.03	19.50	−88.74	97.84
L2KD	32.75	−54.67	78.49	63.17	17.27	38.09
LAMOL_KD	29.96	−71.79	89.10	48.10	−56.19	92.72
PCLL	28.46	−74.38	93.88	48.84	−55.42	92.89
Base + JLT	51.29	−36.30	79.57	73.06	−13.23	79.85
L2KD + JLT	95.35	−1.00	94.38	93.82	−0.16	92.79
LAMOL_KD + JLT	94.72	−0.99	92.16	90.68	−1.03	89.74
PCLL + JLT	95.42	−0.97	94.94	92.24	−0.10	91.98
Qwen3-0.6B						
Joint Fine-tuning	95.65	–	–	96.50	–	–
SEQ*	60.77	0.00	0.00	70.41	0.00	0.00
KLDA-E	94.33	0.00	0.00	94.21	0.00	0.00
Base	6.98	−79.11	73.01	19.99	−89.91	89.94
L2KD	32.90	−58.13	73.79	62.32	9.90	44.41
LAMOL_KD	26.45	−79.12	95.30	42.09	−60.48	93.50
PCLL	31.29	−74.50	96.14	49.40	−55.33	92.50
Base + JLT	52.55	−35.27	78.87	71.03	−18.32	82.20
L2KD + JLT	95.58	−1.15	94.98	92.16	−1.21	91.32
LAMOL_KD + JLT	95.01	−1.87	94.67	91.47	−1.53	90.21
PCLL + JLT	94.96	−1.13	96.55	89.56	−1.75	90.11

Table 18: Stream-level view of the JLT decomposition, computed from Tables 16 and 17. **Top:** residual gap $A_T^{\text{joint}} - A_T^{\text{JLT}}$ per benchmark stream (negative values: JLT slightly exceeds the joint reference). **Bottom:** mean change in operational FWT after task-adaptive head re-alignment (JLT minus standard), averaged over the three backbones.

IL method	Backbone	TACRED	Banking77	CLINC150	FewRel	Topic3
Base	Pythia-410M	0.06	20.59	23.86	44.04	23.29
	Llama-3.2-1B	−0.30	16.79	22.42	44.36	22.44
	Qwen3-0.6B	1.17	18.10	19.42	43.10	25.47
L2KD	Pythia-410M	0.95	0.43	0.13	0.17	1.59
	Llama-3.2-1B	1.27	0.16	0.97	0.30	1.68
	Qwen3-0.6B	0.45	−1.57	1.39	0.07	4.34
LAMOL_KD	Pythia-410M	0.75	0.63	−0.05	−0.38	1.50
	Llama-3.2-1B	1.18	0.90	9.49	0.93	4.82
	Qwen3-0.6B	0.78	1.28	6.19	0.64	5.03
PCLL	Pythia-410M	2.29	2.35	0.06	0.65	2.04
	Llama-3.2-1B	2.03	1.94	1.04	0.23	3.26
	Qwen3-0.6B	2.43	0.70	1.44	0.69	6.94

	Mean ΔFWT (JLT − standard)	
IL method	TACRED + FewRel	Banking77 + CLINC150 + Topic3
Base	−35.40	−2.47
L2KD	−29.51	+21.49
LAMOL_KD	−40.35	−4.30
PCLL	−34.10	−1.06

Table 19: Hyperparameters for LM-head adaptation.

Hyperparameter	Default
LM-head learning rate	5×10^{-4}
Optimizer	AdamW
Loss	Label-token cross-entropy
Epochs (E)	3
Batching	Same as the backbone training loop

Label tokenization. For each instance, we tokenize the gold textual label and take the first token id as target:

$$y^{\text{tok}} = \text{Tokenizer}(\text{label})[0].$$

Given hidden feature $\mathbf{h} \in \mathbb{R}^d$ from the backbone and output embedding $W \in \mathbb{R}^{V \times d}$, the vocabulary logits are

$$\mathbf{z} = W\,\mathbf{h} \in \mathbb{R}^V.$$

Base LM-head loss.

$$\mathcal{L}_{\text{LM}} = \text{CE}\big(\mathbf{z},\, y^{\text{tok}}\big)\,.$$

Equivalently, for task t with N_t training examples, hidden features $\mathbf{h}_t^i = f_T(x_t^i)$ from the frozen backbone after IL, and the label-token mapping $\tau(y) = \text{Tokenizer}(y)[0]$, the task-level head-only objective is

$$\mathcal{L}_t = -\frac{1}{N_t} \sum_{i=1}^{N_t} \log \text{softmax}(W\mathbf{h}_t^i)\big[\tau(y_t^i)\big].$$

Optimization. The LM head parameters are optimized by AdamW with learning rate η (5×10^{-4} by default). During JLT, only the LM-head parameters are passed to this optimizer; the backbone receives no gradient updates.

Table 20: Deployable post-hoc JLT on Llama-3.2-1B. The head is re-aligned once per checkpoint with a small replay buffer of roughly one example per class and restored after evaluation; no task-specific adaptation is performed. Numbers are standard CIL metrics.

		A_T		BWT		FWT		
Dataset	**Method**	Standard	+JLT	Standard	+JLT	Standard	+JLT	Δ
Banking77	Base	26.98	50.55	−82.35	−26.29	95.80	71.29	−24.51
CLINC150	Base	47.56	67.49	−55.36	−17.98	89.86	79.12	−10.74
TACRED	DER++	35.44	59.13	−69.18	−13.34	95.43	64.71	−30.72
TACRED	Base	33.68	61.69	−72.07	−10.79	96.22	67.46	−28.76
Average		35.92	59.72	−69.74	−17.10	94.33	70.65	−23.68

F.1 Evaluation-time LM-head finetune

Before evaluation on task t, the backbone is frozen and the LM head is adapted for E epochs (1–3; 3 by default, Table 19) on the training split of task t using $\mathcal{L}_{\mathrm{LM}}$. Evaluation itself is unchanged: the model generates the label sequence, and a prediction is counted as correct only if it exactly matches the gold label. After evaluation, LM-head weights are restored to their original state to avoid leakage across tasks. Because this protocol uses task-specific training data at evaluation time, it is reported only as a diagnostic estimate of recoverable head performance.

F.2 Post-hoc JLT

Post-hoc JLT is a deployable variant under the same rehearsal assumption as ER, namely that a small number of examples from previously seen tasks can be accessed. After training on task k and before evaluating checkpoint k, the LM head is re-aligned once; after evaluation, the original head is restored exactly. For each seen task $0, \ldots, k$, the buffer takes as many examples as the task has classes from its shuffled training loader (batch size 4), rounded up to whole batches. This gives 8, 12, and 12 examples per task for TACRED (5 classes per task), Banking77 (11), and CLINC150 (10), and 64, 84, and 180 examples in total at the final checkpoint. Unlike task-adaptive JLT, post-hoc JLT never accesses the full training split of an evaluated task and never sees data from future tasks, so its A_T, BWT, and FWT are standard CIL quantities. The buffer size is thus comparable to ER's one example per class, and the buffer is redrawn from the training loaders at each checkpoint.

Only the output embedding (LM head) is trained; the backbone is kept in evaluation mode without gradients. The loss is cross-entropy computed only at the answer-token positions (the gold label and end-of-sequence tokens), optimized with AdamW (learning rate 5×10^{-4}, weight decay 0.01, no scheduler, no gradient clipping) for 3 epochs with batch size 4. Experiments use Llama-3.2-1B in bf16 with FlashAttention-2, seed 42, and 5 IL training epochs per task. Table 20 reports the results. Post-hoc JLT uses a separate set of Llama-3.2-1B runs, so its standard baselines differ from those in Tables 16 and 17.

G Comparison with prior analyses

Table 21 compares our framework with prior analyses of representation forgetting. Prior work mainly measures stability and shows that much previous-task information remains in the backbone while the head degrades; it does not quantify representation-level plasticity or compare the BWT–FWT relation across interfaces.

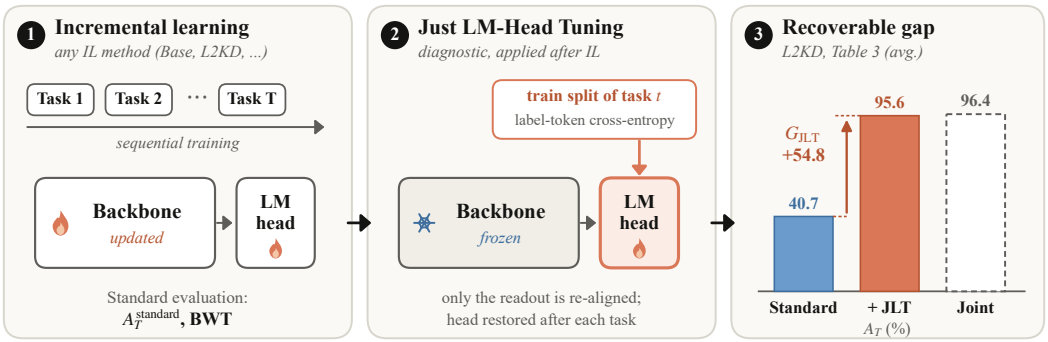


Figure 11: Overview of JLT as a head-only diagnostic. (1) The IL process follows the basic procedure of each IL method, updating the backbone and LM head task by task. (2) After IL training, the backbone is frozen and only the LM head is re-aligned with a label-token cross-entropy objective. (3) The resulting gap G_{JLT} estimates how much performance is recoverable through head re-alignment; the bars show the L2KD averages from Table 3 together with the joint fine-tuning reference.

Table 21: Comparison with prior analyses of forgetting at the representation level.

Work	**Interfaces**	**BWT**	**FWT / plasticity**	**Classifier-free**	**Generative LM head**	**Interface-wise BWT–FWT**
Davari et al. [2022]	Head, probe	✓	–	–	–	–
Zhou and Srikumar [2021]	DirectProbe	–	–	Partial	–	–
Zheng et al. [2024]	Head, probe	✓	–	–	Partial	–
Zheng et al. [2025a]	Head	✓	–	–	✓	–
Chen et al. [2023]	Head	✓	Head only	–	–	–
Ours	Head, probe, backbone, JLT	✓	Representation level	✓	✓	✓

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